

B-fields In Star-forming Region Observations (BISTRO): Dual-Wavelength Constraints on Size-Dependent Grain Alignment in Star-Forming Regions

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ABSTRACT

We report three systematic findings from the BISTRO dual-wavelength (450/850 μm) polarimetric survey, using metric-specific high-quality subsamples drawn from the broader survey: the 21-source dual-band-angle sample for the polarization-angle comparison and the 28-source SPIDX subsample for the spectral-index analysis (short-name convention adopted across this paper series; see companion Paper S2 for the complete 69-region POLSRC census and Paper T Table 2 for the full seven-sample inventory). (F1) Magnetic field orientations agree within $|\chi_{450} - \chi_{850}| \lesssim 10^\circ$ across the 21-source dual-band-angle sample ($N_{\text{samp}} = 1649$), consistent with the post-harmonization polarization-angle floor of $|\Delta\chi| \sim 4^\circ\text{--}6^\circ$ established in companion Paper T §3.5, ruling out line-of-sight geometric depolarization as the dominant origin of the wavelength-dependent polarization. (F2) The polarized-intensity spectral index γ is systematically smaller than the total-intensity spectral index α in 21 of 28 sources (75%); the per-source medians are $\langle\alpha\rangle = 3.46$ and $\langle\gamma\rangle = 3.12$, producing a representative per-source difference $\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ (systematic floor of companion Paper T §4.2 and Table 6; statistical empirical standard error of the median estimated from the dispersion per-source of the SPIDX sample). (F3) The polarization-fraction ratio $P_{450}/P_{850} < 1$ shows no dependence on the inter-band polarization-angle difference $|\chi_{450} - \chi_{850}|$, isolating intrinsic grain physics as the origin rather than geometric artifacts; the sample median is $P_{450}/P_{850} = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$ (empirical), with values per-source spanning the range of 0.4–1.4. Taken together, these findings isolate the origin of the polarization spectrum and favor size-dependent alignment, in which the grains dominating the polarized emission are aligned more efficiently and exhibit a shallower effective emissivity spectrum than the bulk dust population. The resulting quantitative benchmarks— $\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ and $P_{450}/P_{850} = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$ (per-source range 0.4–1.4)—constrain dust models and magnetohydrodynamic simulations. The *existence-level* findings F1–F3 reported here are complemented by the *universality-level* findings F4–F6 reported in companion Paper S2: F4 confirms F1–F3 as population-level phenomena across the BISTRO POLSRC sample, F5 demonstrates regime-dependent wavelength response, with the median $\gamma - \alpha$ and R_p both increasing from LMSF through HMSF to CMZ, and F6 quantifies disorder-mediated wavelength divergence under local N_{H} and T_d coupling within the Herschel Gould Belt Survey 9-field subsample. More broadly, the three column-independent observables introduced here — P_{450}/P_{850} , $\gamma - \alpha$, and $|\chi_{450} - \chi_{850}|$ — define a practical diagnostic framework for submillimeter emission polarimetry, analogous to the role of p_V/A_V in extinction studies, and provide a standardized basis for future multi-wavelength submillimeter polarization surveys.

Keywords: Interstellar magnetic fields (845) — Submillimeter astronomy (1647) — Dust physics (2229)
 — Polarimetry (1278) — Star formation (1569) — Interstellar dust (836) — James Clerk
 Maxwell Telescope (2109)

1. INTRODUCTION

Magnetic fields pervade the interstellar medium and play a central role in the dynamics of molecular cloud formation, filament evolution, and star formation (Hennebelle & Inutsuka 2019; Pattle et al. 2023). Submillimeter polarimetry of thermal dust emission has emerged as the primary tool for mapping plane-of-sky magnetic field morphology in star-forming regions, complementing Zeeman measurements of the line-of-sight component (Crutcher 2012). Recent observations have established that plane-of-sky field strengths in nearby molecular clouds span roughly three orders of magnitude, from $\sim 1 \mu\text{G}$ in diffuse envelopes to $\sim 10 \text{ mG}$ in the densest star-forming cores such as the OMC 1 integral-shaped filament where Davis–Chandrasekhar–Fermi analysis yields $\sim 6 \text{ mG}$ (Pattle et al. 2017, 2022, 2024), with the *Planck* all-sky survey and *Herschel*-based column density maps providing the large-scale environmental context for interpreting these results (Planck Collaboration et al. 2015; André et al. 2010).

Physical basis.—Submillimeter emission polarimetry probes magnetic fields in cold ($T \lesssim 100 \text{ K}$) gas through the alignment of non-spherical dust grains with their major axes perpendicular to the local B -field direction (Davis & Greenstein 1951). Thermal emission from such aligned grains is polarized, enabling inference of the plane-of-sky B -field orientation. These dust grains—a mixture of silicates and carbonaceous species comprising $\sim 1\%$ of the ISM by mass (Draine 2003)—emit optically thin radiation whose intensity traces the column density (Hildebrand 1983).

Observational context.—The JCMT POL-2 instrument (Friberg et al. 2016), operating with the SCUBA-2 camera (Holland et al. 2013), provides wide-field dual-wavelength ($450/850 \mu\text{m}$) submillimeter polarimetry. Its high dynamic range enables polarization measurements across hydrogen column densities of $N_{\text{H}} \sim 10^{20}\text{--}10^{24} \text{ cm}^{-2}$, and its $\sim 3'$ field of view enables efficient mapping of extended structures. The ongoing B-fields In STar-forming Region Observations (BISTRO) Large Program (Ward-Thompson et al. 2017) has systematically mapped magnetic field morphology in diverse star-forming regions, but $450 \mu\text{m}$ observations remain challenging due to atmospheric variability; the methodological advances that establish, validate, and quantify the systematic floors of reliable $450 \mu\text{m}$ polarimetry are presented in companion Paper T.

Extinction polarimetry versus emission polarimetry.—A conceptual difference between optical and near-infrared extinction polarimetry and submillimeter emission polarimetry deserves emphasis. In extinction polarimetry, the polarization fraction p_V and extinction A_V both scale with the column of dust along the line of sight, so their ratio p_V/A_V naturally cancels the dust column and isolates the intrinsic polarizing efficiency of the grain population—a quantity traditionally termed “polarization efficiency” (Serkowski et al. 1975; Whittet et al. 2008). In emission polarimetry, by contrast, the polarized intensity PI_ν and the total intensity I_ν both carry a temperature-dependent weighting through the Planck function $B_\nu(T)$. Because different grain sizes equilibrate at different temperatures and contribute differently at different wavelengths, the ratio $P_\nu = PI_\nu/I_\nu$ conflates alignment efficiency with grain optical properties and temperature structure along the line of sight. Consequently, no simple analog of p_V/A_V exists for submillimeter emission polarimetry that cleanly isolates alignment efficiency from other factors.

Grain-alignment background.—The most widely accepted theory for dust grain alignment is the Radiative Alignment Torque (RAT) mechanism (Lazarian 2007; Lazarian & Hoang 2007a), in which irregularly shaped dust grains are spun up by anisotropic radiation fields (Andersson et al. 2015).

The polarization hole paradigm.—Because the RAT mechanism aligns grains through incident radiation, alignment efficiency should decrease as extinction increases (Lazarian 2007; Andersson et al. 2015). This trend manifests itself as the *polarization hole*: the polarization fraction P decreases with increasing column density N . The phenomenon was first systematically characterized by Goodman et al. (1995) and has been confirmed in near-infrared extinction polarimetry (Whittet et al. 2008; Alves et al. 2014) and submillimeter emission polarimetry (Kwon et al. 2018; Soam et al. 2018; Coudé et al. 2019).

Internal heating as a competing effect.—However, the simple polarization hole paradigm may not apply universally. In star-forming regions containing embedded protostars, locally generated radiation fields can exceed the ISRF by orders of magnitude (Whitney et al. 2003; Robitaille et al. 2006). These enhanced internal radiation fields may drive efficient grain alignment even at high column densities, potentially reversing the classical negative P – N correlation. Whether the P – N correlation is positive or negative in actively star-forming regions—and whether this behavior differs between wavelengths—remains an open question that our dual-wavelength survey is positioned to address.

Size-dependent grain alignment.—RAT theory makes a fundamental prediction: alignment efficiency increases strongly with grain size, because larger grains experience greater radiative torques relative to gas damping (Lazarian & Hoang 2007a; Hoang & Lazarian 2016). This size dependence has profound implications for multiwavelength polarimetry. If large grains align more efficiently than small grains, and if different grain sizes contribute preferentially at different wavelengths, then polarization properties should exhibit a systematic wavelength dependence.

Laboratory measurements and theoretical models indicate that grain coagulation and mantle growth can flatten the far-infrared opacity spectrum of large composite aggregates relative to bare silicates (Köhler et al. 2015; Ysard et al. 2019). If polarized emission is dominated by well-aligned large grains exhibiting a shallower effective emissivity spectrum, then the polarization-weighted spectral index should differ systematically from the mass-weighted spectral index—a testable prediction that has not been systematically examined across diverse star-forming environments.

The polarization spectrum.—Simple dust models predict that submillimeter polarization fraction varies only weakly with wavelength (Hildebrand et al. 1999), but observations have revealed a complex polarization spectrum with structure between 60 μm and 1.3 mm (Vaillancourt et al. 2008; Gandilo et al. 2016; Shariff et al. 2019). The THEMIS 2.0 dust model predicts polarization spectrum slopes of -0.5 to -1.0 between 450 and 850 μm for composite grains with iron inclusions (Ysard et al. 2024).

Prior multiwavelength studies.—Fanciullo et al. (2022) combined Stratospheric Observatory for Infrared Astronomy (SOFIA)/HAWC+ (154, 214 μm) and JCMT/POL-2 (850 μm) observations of NGC 2071, finding wavelength-dependent polarization fractions attributed to grain alignment dependent on size. Michail et al. (2021) provided complementary constraints in Orion using HAWC+ polarimetry. However, these studies have focused on individual sources; a systematic survey across diverse environments is needed to establish whether the observed behaviors are universal or source-specific.

The approach of this study.—This study addresses the interpretive challenge of submillimeter emission polarimetry by employing three observational diagnostics that isolate size-dependent grain alignment without requiring explicit normalization by dust column: the polarization fraction ratio P_{450}/P_{850} , which compares alignment efficiency between grain populations dominating at different wavelengths; the spectral index difference $\gamma - \alpha$, which contrasts the wavelength dependence of polarized and total intensity; and the P – $\mathcal{S}$ correlation between the polarization fraction P and the polarization-angle dispersion function $\mathcal{S}$, which controls for geometric depolarization from magnetic-field tangling. By examining how these diagnostics vary across diverse environments, we extract constraints on size-dependent alignment efficiency that are independent of the absolute dust column or temperature structure. This approach parallels the diagnostic role that p_V/A_V has played in extinction polarimetry, providing column-independent metrics that constrain the intrinsic physics of grain alignment across the grain size distribution. In this sense, the present work is not only a survey analysis but also an attempt to formalize a set of wavelength-differential observables that are intrinsically less sensitive to absolute column normalization than conventional polarization fraction measurements alone; we suggest that these quantities may serve as a standardized observational toolkit for emission polarimetry in the same way that extinction-normalized diagnostics became foundational in optical and near-infrared grain-alignment studies.

Open questions and survey objectives.—Despite these observational and theoretical advances, fundamental questions remain unanswered:

First, does submillimeter polarimetry trace a single plane-of-sky magnetic field geometry across wavelengths, or do different bands sample different magnetized structures? In the 450/850 μm comparison specifically, this reduces to whether the two bands yield consistent polarization angles. Establishing a wavelength-independent field orientation would confirm that any differences in polarization fraction arise from intrinsic grain properties rather than geometric effects.

Second, how does the spectral index of polarized emission compare with that of total emission across diverse star-forming environments? In the 450/850 μm comparison specifically, this contrasts the polarized-intensity index γ with

the total-intensity index α . The prediction that well-aligned large grains exhibit a shallower effective emissivity spectrum than the bulk grain population implies $\gamma < \alpha$ —a relationship that has not been systematically tested.

Third, is the wavelength dependence of the submillimeter polarization fraction set by intrinsic grain physics, or is it shaped by line-of-sight magnetic-field tangling? In the 450/850 μm comparison specifically, this asks whether P_{450}/P_{850} correlates with the inter-band polarization-angle difference $|\chi_{450} - \chi_{850}|$, where χ_λ denotes the plane-of-sky polarization angle measured at wavelength λ . If $P_{450}/P_{850} < 1$ reflects intrinsic grain physics, no such correlation should appear.

Using the statistical subsample drawn from the broader 69-source survey (companion Paper S2), we address these three questions through systematic dual-band analysis at 450 and 850 μm .

To address these three questions, the analysis requires dual-band data that are co-spatial, consistently reduced, and sufficiently sensitive at 450 μm to permit matched-resolution comparisons of polarization fraction, spectral indices, and magnetic-field orientation. Section 2 therefore focuses on the sample definition and data products needed to construct those diagnostics.

Structure of this paper.—Section 2 describes the observations and data. Section 3 presents the three findings: B-field angle consistency (§3.1; F1), polarization-fraction ratios (§3.2; F3), and spectral-index analysis (§3.3; F2). Section 4 develops the physical interpretation in terms of size-dependent grain alignment, environmental response, and implications for RAT theory. Section 5 summarizes the three principal findings. The distance determination methodology is described in companion Paper S2 Appendix E; Appendix A assesses $^{12}\text{CO}(J=3-2)$ contamination. The companion technical paper, Paper T, establishes the noise characterization, demonstrates the 450 μm offset correction—which is independently validated against SOFIA/HAWC+ at 154 μm —and quantifies the systematic-uncertainty floors that the dual-band measurements used here rest upon. The companion census paper (Paper S2) extends the F1–F3 existence layer presented here to a unified F1–F6 program: F4 confirms population-level universality across the BISTRO POL-SRC sample, F5 demonstrates regime-dependent wavelength response, with the median $\gamma - \alpha$ and R_p both increasing from LMSF through HMSF to CMZ, and F6 quantifies disorder-mediated wavelength divergence within the Herschel Gould Belt Survey 9-field subsample.

For navigation, the present paper provides two complementary architectural aids. Figure 1 summarises the three-finding framework as a schematic in the question-to-finding direction. Table 1 provides the inverse-direction cross-index (technical topic to F1, F2, and F3 mapping), enumerating each finding and indicating its location together with the relevant companion-paper anchor (Paper T Pillar that underpins it, or Paper S2 finding that extends it).

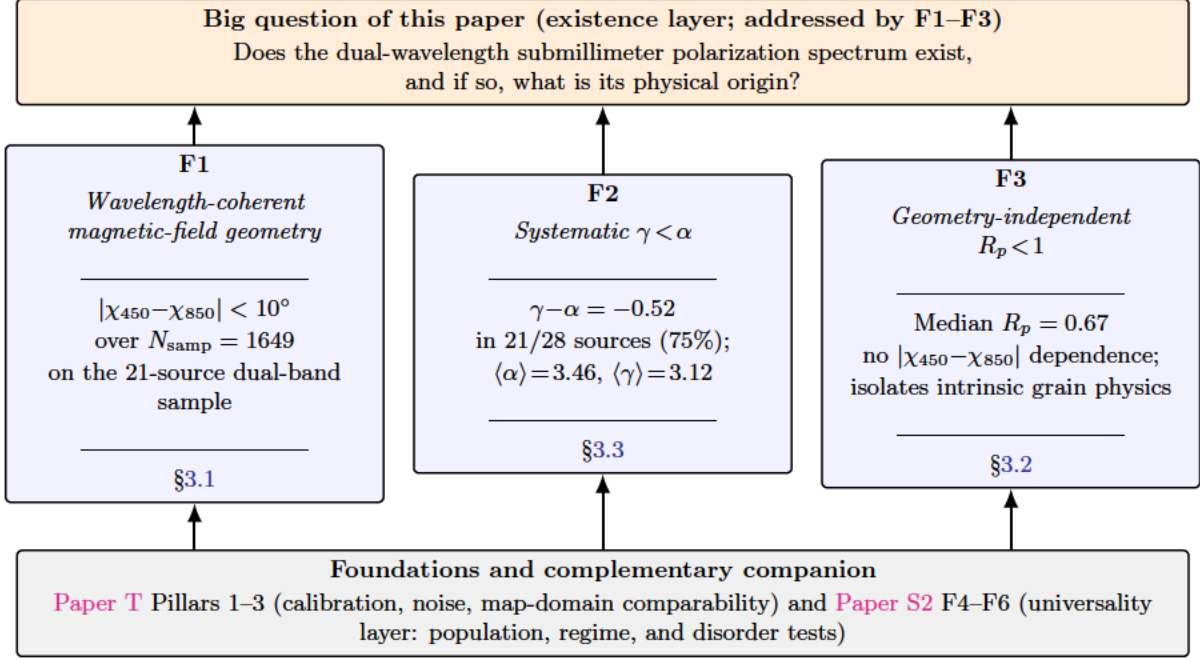


Figure 1. Architectural overview of Paper S1’s three-finding framework (existence layer of the unified F1–F6 program). The big question of this paper—existence and physical origin of the dual-wavelength submillimeter polarization spectrum—is addressed through three complementary observational findings on a SPIDX subsample of the BISTRO survey. *F1* (*wavelength-coherent magnetic-field geometry*): per-pixel agreement $|\chi_{450} - \chi_{850}| < 10^\circ$ over $N_{\text{samp}} = 1649$ matched-resolution samples rules out line-of-sight geometric depolarization as the dominant origin of the wavelength-dependent polarization spectrum (§3.1). *F2* (*systematic $\gamma < \alpha$*): the polarized-intensity spectral index γ is smaller than the total-intensity spectral index α in 21 of 28 sources (median $\gamma - \alpha = -0.52$), interpreted as a lower effective emissivity index β for the polarization-weighted grain population (§3.3). *F3* (*geometry-independent $R_p < 1$*): the polarization-fraction ratio $R_p \equiv P_{450}/P_{850} = 0.67$ shows no dependence on the inter-band polarization-angle difference, isolating intrinsic grain physics rather than geometric artefacts as the origin (§3.2). The three findings rest on the technical credibility chain of companion Paper T (Pillars 1–3), and are extended to the universality layer (F4–F6) by companion Paper S2.

Table 1. Cross-reference index of Paper S1’s three-finding framework, with cross-references to companion-paper material

Topic	Finding	Location in this paper series	Key result / figure	Note
<i>Paper S1 — Existence-layer findings (F1–F3)</i>				
B-field angle agreement	F1	§3.1	$ \chi_{450} - \chi_{850} < 10^\circ$, $N_{\text{samp}} = 1649$ on 21-source sample	1
Polarization-fraction ratio R_p	F3	§3.2	sample-median $R_p = 0.67$, no $ \Delta\chi $ dependence	2
Spectral indices α , γ	F2	§3.3	$\gamma - \alpha = -0.52$ in 21/28 sources (75%)	3
<i>Paper S1 — Discussion synthesis</i>				
Three-findings synthesis	—	§4	size-dependent grain alignment (RAT-consistent)	4
<i>Companion-paper material (cross-referenced from Paper S1)</i>				
Pillar 1 (FCF + Planck calibration)	—	Paper T §3.1	underpins α , γ used in F2	5
Pillar 2 (Stokes $I + Q, U$ noise model)	—	Paper T §§3.2–3.3	underpins per-pixel detection masks used in F1, F3	6
Pillar 3 (offset corr. + beam harmonization)	—	Paper T §§3.4–3.5	underpins matched-resolution maps used throughout	7
F4: Population-level universality of F1–F3	—	Paper S2	POLSRC sample confirms F1–F3 are sample-wide	8
F5: Regime-dependent wavelength response	—	Paper S2	$\gamma - \alpha$, R_p increase from LMSF to CMZ	9
F6: Disorder-mediated wavelength divergence	—	Paper S2	HGBS 9-field per-pixel N_H, T_d correlations	10

NOTE— “Finding” column lists S1-specific findings (F1–F3); F4–F6 reside in Paper S2 and are cross-referenced. “Topic” column lists scientific items in the order they appear in the text. “—” in the Finding column indicates synthesis material or companion-paper cross-references rather than a primary S1 finding. (1) Per-pixel polarization-angle agreement on the 21-source dual-band sample at $I/\sigma_I > 100$; basis for the 10° physical-tolerance scale used throughout. (2) Three statistical estimators (Methods 1, 2, 3 of Paper S2 §3.3) are mutually consistent; Method 3 (median of pixel ratios) is canonical for the per-source census. (3) Per-source spectral indices on the SPIDEX subsample; Planck shape factor $C(T_d)$ cancels in the difference $\gamma - \alpha$. (4) Discussion synthesis develops the size-dependent grain alignment interpretation: aligned large grains exhibit a shallower effective emissivity spectrum than the bulk dust population, jointly accounting for F2 ($\gamma < \alpha$) and F3 ($R_p < 1$) under the geometry-independence guarantee of F1. (5) FCF_{arcsec} choice and Planck-corrected $\beta = \alpha - C(T_d)$ framework underpin the α , γ , β definitions used throughout F2. (6) Two-component noise model and isotropy verification underpin the per-pixel detection masks used in F1 angle agreement and F3 polarization-fraction ratio. (7) Three-step PCA-based offset correction and beam harmonization underpin the matched-resolution Stokes maps that feed all three findings. (8) Per-source measurements across the BISTRO POLSRC sample test whether F1–F3 are sample-wide phenomena rather than artefacts of representative sources; sample medians confirm $R_3 = 0.67$, $\gamma - \alpha = -0.52$, $|\chi_{450} - \chi_{850}| < 10^\circ$ universally. (9) Stratification across LMSF/HMSF/CMZ sub-environments reveals systematic monotonic ordering of $\gamma - \alpha$ and R_p while the F1 wavelength-coherence floor is preserved in every environment. (10) Per-pixel correlations of polarization observables with HGBS-derived N_H, T_d in 9 fields demonstrate that the wavelength dependence couples preferentially to local conditions at $450 \mu\text{m}$.

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Table 2 provides the per-finding quantitative master view of F1–F3; the comprehensive cross-reference index of this paper’s organisation appears as Table 1 above.

Table 2. Paper S1 F1–F3 quantitative master summary.

Label	Observable	Sample	Result	Stat. unc.	Sys. floor	Paper-of-record	Limitations
F1	$ \chi_{450} - \chi_{850} $ inter-band agreement	21-source dual-band ($N_{\text{samp}} = 1649$)	$ \widetilde{\Delta\chi} < 10^\circ$ all-source (Mon R2 $\sim 11^\circ$)	per-pixel $\sigma_{\Delta\chi} \approx 1.1^\circ$	4–6° floor (T §3.5)	§3.1	Mon R2 joint stress test
F2	$\gamma - \alpha$ spectral-index difference	28-source SPIDX	$\gamma - \alpha = -0.52$, 21/28 sources $\gamma < \alpha$	empirical SEM ~ 0.10 per-source	$\lesssim 0.04$ (T Table 6)	§3.3	T-mixing same-component scope (§4.4)
F3	$R_p \equiv P_{450}/P_{850}$ ratio	28-source SPIDX + HGBS-9 (417 pix)	$R_3 = 0.67$ sample median	empirical SEM ~ 0.04 per-source	$\lesssim 0.02$ (T Table 6)	§3.2	OMC-1 high-radiation regime cancellation in R_p

NOTE— The three rows summarise the home findings of this paper in the eight-column quantitative format introduced for cross-comparison with the F4–F6 master table of companion **Paper S2 Table 2**. Statistical uncertainties follow the median $\pm$ SE-of-median convention ($\text{SE} = 1.253 \text{ MAD}/\sqrt{N}$) adopted across the paper series; “empirical SEM” indicates per-source dispersion-based estimates used where a closed-form SE is unavailable. Systematic-floor entries are imported from the technical-precision framework of companion **Paper T** as catalogued in Table 3. The legacy comprehensive cross-reference index of Table 1 retains the full row inventory including discussion-synthesis and companion-paper-material entries; the present table focuses on the per-finding quantitative content.

The three-paper context.—These three papers together address the broader question of what governs the wavelength dependence and environmental manifestation of submillimeter dust polarization in star-forming regions: [Paper T](#) establishes the methodological observability framework, the present paper ([Paper S1](#)) identifies the universal wavelength-dependent signatures, and [Paper S2](#) examines whether and how those signatures reorganize across environments.

2. OBSERVATIONS AND DATA

Data and sample.—The data used in this study were selected from POL-2 observations available in the JCMT archive as of 2022 February. We limited the analysis of 450 μm data to observations taken under the highest-quality Weather Band (WB 1), corresponding to zenith-atmospheric opacity $\tau_{225\text{ GHz}}^{\text{zenith}} \leq 0.05$. Because SCUBA-2 simultaneously acquires data at both 450 and 850 μm in every observation, the dual-band dataset is inherently co-spatial and co-temporal, eliminating cross-calibration uncertainties between wavelengths. The resulting sample comprises 69 distinct star-forming regions (the POLSRC sample, in the short-name convention adopted across this paper series); the complete sample inventory (with per-source coordinates, distances, and archival POL-2 references) is given in [Paper T Table 2](#).

Sample selection function.—Our sample selection is governed by three criteria: availability to the lead author as of 2022 February, 450 μm observations obtained under Weather Band 1 conditions, and sufficient signal-to-noise ratio for dual-band polarization analysis (peak Stokes I signal-to-noise ratio > 100 at 450 μm). These requirements inherently bias the sample toward high surface-brightness, nearby star-forming regions with strong continuum emission. Consequently, the statistical results presented in this paper characterize the polarization properties of the brightest subregions within molecular clouds—predominantly dense cores and filaments actively forming stars—rather than representing a volume-complete or flux-limited census of star-forming environments in the solar neighborhood. This selection function necessarily favors relatively bright, nearby star-forming cloud interiors in which the 450 μm polarized signal is measurable at matched resolution with the 850 μm data. The resulting statistical sample should therefore be regarded as representative of the high-surface-brightness regime currently accessible to sensitive dual-band submillimeter polarimetry, rather than of the full dynamic range of molecular-cloud environments. More diffuse or weakly illuminated regions are observationally underrepresented, because their shorter-wavelength polarized emission falls below the practical sensitivity floor of the present JCMT/POL-2 survey. This limitation should be kept in mind when generalizing the physical trends derived below.

Data reduction.—Both wavelengths were reduced using the `pol2map` pipeline within `Starlink` (Chapin et al. 2013), following the validated three-step procedure established for POL-2 data (e.g., Ward-Thompson et al. 2017; Pattle et al. 2021). To ensure that the 450 and 850 μm maps sample identical spatial scales, we adopted `flt.filt.edge.largescale=300`, which retains structure smaller than $\sim 300''$, together with `smooth450=yes`, which deconvolves the double-Gaussian 450 μm beam and reconvolves to the $12''.6$ effective 850 μm beam; the common sensitivity range is thus $12''.6\text{--}300''$. All Stokes parameters were regridded onto a common coordinate system with $12''$ pixels and matched to the 850 μm beam resolution for dual-band analysis. Flux calibration, surface-brightness unit conversion, noise characterization, the 450 μm offset-correction procedure with independent SOFIA/HAWC+ cross-validation at 154 μm , and the complete systematic-uncertainty budget are documented in the companion Paper T (§§3–8); in the present paper the 450 μm offset-correction residual enters only through the systematic floor δX_{sys} in the unified uncertainty notation introduced at §3. The debiased polarization fractions were calculated using the modified asymptotic estimator (Plaszczynski et al. 2014; Montier et al. 2015); implementation details are given in Paper T Appendix A. Potential $^{12}\text{CO}(J=3\text{--}2)$ contamination of the 850 μm continuum band is assessed in Appendix A and is found not to affect the principal polarization results at the level relevant for the present analysis.

Distance assignment.—Distances were assigned to all survey fields using a hierarchical approach based on Gaia-derived catalogs, supplemented by literature values for well-studied sources; the detailed methodology is presented in companion [Paper S2 Appendix E](#). These distances should be regarded as reference scaling factors to convert angular measurements to physical units, rather than as precise physical distances, given the intrinsic line-of-sight depth of the spatially extended clouds studied here. Importantly, the primary analyses presented in this paper are intentionally constructed to be largely insensitive to distance uncertainties: they rely predominantly on ratio-based or angular quantities— P_{450}/P_{850} , $\gamma - \alpha$, $|\chi_{450} - \chi_{850}|$, and rank-based correlation statistics—for which distance-dependent scaling factors either cancel or do not enter.

3. RESULTS AND ANALYSIS

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Table 3 summarises the technical primitives imported from companion Paper T and the F4–F6 population-level findings imported from companion Paper S2 that the F1–F3 analyses of this section build upon.

Table 3. Paper S1 imports from companion papers (foundation for the F1–F3 analyses of §3)

Imported framework / finding	Source	Used in this paper
Extended-source FCF + Planck-corrected β (Pillar 1)	Paper T § 3.1	§3.3 (α, γ definitions for F2)
Stokes- I + Q, U analytic noise model (Pillar 2)	Paper T §§ 3.2–3.3	per-pixel detection masks for F1 (§3.1) and F3 (§3.2)
450 μm Q/U offset correction + beam harmonization (Pillar 3)	Paper T §§ 3.4–3.5	matched-resolution dual-band maps used throughout F1, F2, F3
Systematic-uncertainty floor $\sigma(P_{450}/P_{850}) \lesssim 0.02$, $\sigma(\gamma - \alpha) \lesssim 0.04$	Paper T § 4.1, Table 6	δX_{sys} entries throughout §3
F4–F6 population-level extension	Paper S2 § 3	§4 integrated synthesis

NOTE— The four Paper T dependencies enter through the unified uncertainty notation and the matched-resolution maps used throughout §3; the Paper S2 dependency enters through the integrated grain-alignment discussion (§4). This paper exports the F1–F3 findings developed here to Paper S2, which extends them to the population-level F4–F6 layer; the reverse-direction view is in Paper S2 Table 3.

Notation for uncertainties.—Throughout §3–§5, every flagship physical quantity is presented in a unified form

$$X = \bar{X} \pm \delta X_{\text{sys}} \pm \delta X_{\text{stat}},$$

where $\bar{X}$ is the central value (typically the median per-pixel or per-source, as specified at each point of use), δX_{sys} is the systematic floor inherited from the uncertainty budget of companion Paper T §4.1 (dominated by $450\text{ }\mu\text{m}$ offset-correction residuals; for the principal flagship quantities $\sigma(P_{450}/P_{850}) \lesssim 0.02$ and $\sigma(\gamma - \alpha) \lesssim 0.04$), and δX_{stat} is the statistical scatter on the central value (standard error of the median over the relevant pixel or source ensemble). The two budgets are reported separately because they probe distinct sources of uncertainty: δX_{sys} measures how confidently the central value is anchored to the absolute calibration, while δX_{stat} measures how reproducibly it is recovered within our sample. Where a closed-form δX_{stat} is not separately tabulated — typically because the quantity is reported as a single ensemble median rather than as a distribution of per-source estimates — we adopt empirical/typical values inferred from the per-source dispersion of the same ensemble and flag them as such (hereafter “empirical SEM”).

Sample roadmap.—Each of the three findings F1–F3 is driven by a different physical question and therefore a different natural data product; a single one-size-fits-all subsample would either discard usable information or contaminate one test with irrelevant pixels. We therefore adopt a nested family of subsamples of the JCMT BISTRO $450/850\text{ }\mu\text{m}$ POL-2 survey, each tuned to the statistical requirements of a single test. (i) The full 69-field survey underlies the survey-level prevalence statement that $R_p < 1$ in roughly four-fifths of sources (§3.2; per-source census in companion Paper S2 §3.3). (ii) The SPIDX sample (Method-3 pixels with per-source $N_{\text{samp}} \geq 5$; low-mass star-forming (LMSF) / high-mass star-forming (HMSF) / Central Molecular Zone (CMZ) = 9/12/7) underlies the F2 result $\gamma < \alpha$ in 75 % of sources (§3.3, Figure 6). (iii) The 21-source dual-band-angle sample ($N_{\text{samp}} = 1649$ matched-resolution samples with valid χ_{450} and χ_{850} at the matched $12''.6$ resolution) underlies the F1 angle-consistency test (§3.1, Figure 2). (iii') The 13-target per-target subset (the 13 sources within the 21-source dual-band-angle sample with sufficient pixels per source to compute a reliable per-source median $|\chi_{450} - \chi_{850}|$) underlies the per-target inspection of Figure 2(b). (iv) The HGBS 9-region per-pixel subsample (417 Method-3 pixels co-spatial with *Herschel* Gould Belt Survey (André et al. 2010) N_{H_2} and T_d maps) underlies the per-pixel P – S geometric-depolarization control (§3.2, Figure 4) and the I_{450}/I_{850} – R_p anticorrelation (§3.3, Figure 5). Each subsection re-states the relevant subsample selection at its point of use; the per-source tabulations of the 28- and 21-source samples and the spectral-index source-selection criteria are given in companion Paper S2 (§3.1, §3.2).

3.1. *B-field angle consistency*

Notation.—Throughout this paper we distinguish four angle-related quantities.

$\Delta\chi_{\text{tech}}$: Processing-induced polarization-angle residual from the $450\text{ }\mu\text{m}$ beam- and spatial-frequency harmonization procedure; characterized empirically in companion Paper T §3.5, with source-level median $|\Delta\chi_{\text{tech}}| \sim 4^\circ\text{--}6^\circ$ and all sources $< 10^\circ$ at $I/\sigma_I > 30$.

$\Delta\chi_{\text{obs}} \equiv \chi_{450} - \chi_{850}$: Observed per-pixel inter-band polarization-angle difference; used in the dual-band field-geometry tests of this paper and with the same convention in companion Paper S2.

$\sigma_{\Delta\chi}$: Per-pixel formal statistical uncertainty on $\Delta\chi_{\text{obs}}$, propagated from the Stokes- Q, U noise model of companion Paper T §3.3 (survey-median $\sigma_{\Delta\chi} \approx 1.1^\circ$).

10° **tolerance**: Adopted *physical tolerance scale*—a conservative budget for the combined physical and systematic interpretation uncertainty (*not* a measurement error); the components are itemized in the “Physical tolerance scale” paragraph below.

We first verify that the 450 and $850\text{ }\mu\text{m}$ polarization data trace the same magnetic-field structure along each line of sight, a precondition for the per-pixel comparison of the polarization properties presented in §3.2–§3.3. All $450\text{ }\mu\text{m}$ polarization data used here have undergone the three-step offset correction of companion Paper T §3.4, validated against SOFIA/HAWC+ at $154\text{ }\mu\text{m}$ in Paper T §3.4. Figure 2(a) examines the wavelength-independence assumption directly by comparing the polarization angles at 450 and $850\text{ }\mu\text{m}$ for all pixels with valid P_{450}/P_{850} at the matched $12''.6$ resolution ($N_{\text{samp}} = 1649$ across 21 sources). Crucially, this selection applies only signal-to-noise thresholds on the Stokes parameters at each band, with no pre-selection on the angle difference itself, so the observed wavelength

independence is an empirical property of the joint $\chi_{450}-\chi_{850}$ distribution rather than a consequence of the per-band detection mask (or of the narrower Method 3 mask subsequently used in §3.2–§3.3, which is likewise a polarized-intensity S/N threshold with no angle pre-selection). The data clusters tightly along the 1:1 line, with $|\chi_{450} - \chi_{850}| < 10^\circ$ for 76 % of pixels.

Physical tolerance scale.—We adopt 10° as a *physical tolerance scale*, reflecting the characteristic magnitude of systematic effects in the dual-band analysis. These include beam-matching residuals (§2; consistent with the characteristic beam-processing residual floor of $|\widetilde{\Delta\chi}| \sim$ a few-to-several degrees established empirically across the dual-band statistical sample in companion Paper T §3.5, with all sources reaching post-harmonization $|\widetilde{\Delta\chi}| < 10^\circ$ at $I/\sigma_I > 30$ and L1688 contributing the source-level value $|\widetilde{\Delta\chi}| = 4.3^\circ$); residual offset-correction uncertainties from companion Paper T §3.4; spatial-filtering differences between the two bands; and cross-validation against SOFIA/HAWC+ at $154 \mu\text{m}$, which yields typical post-correction offsets of $\sim 8^\circ$ (Paper T §3.4).

Statistical detection limit.—The corresponding *statistical* detection limit derived from the Stokes Q, U noise model of companion Paper T §3.3 is much smaller: the median per-pixel uncertainty on $\Delta\chi$ is $\sigma_{\Delta\chi} \approx 1.1^\circ$, with $3\sigma_{\Delta\chi} \approx 3.4^\circ$. Pixel-to-pixel deviations within the 10° physical tolerance can therefore be statistically significant even though they remain physically small compared with the full 180° angular range of polarization. In the unified notation introduced at the opening of this section, a typical wavelength-independent pixel is therefore characterized by $\Delta\chi \equiv \chi_{450} - \chi_{850} = \bar{\Delta\chi} \pm 10^\circ_{\text{sys}} \pm 1.1^\circ_{\text{stat}}$, with $\bar{\Delta\chi} \lesssim 1^\circ$ for the bulk of the sample; the systematic component is the physical tolerance budget (above) and the statistical component is the median formal uncertainty per-pixel (Paper T §3.3).

Per-target check and Monoceros R2 outlier.—Per-target inspection (Figure 2b) confirms that this PA agreement holds for nearly all sources individually: the median $|\chi_{450} - \chi_{850}|$ falls below the 10° criterion for 12 of 13 targets. The single exception is Monoceros R2, for which the median $|\chi_{450} - \chi_{850}| = 11^\circ$ slightly exceeds this threshold (16th–84th percentile range 4.5° – 25°), with a systematic signed offset of $+8.8^\circ$.

This behavior is consistent with the previously noted peculiarity of Monoceros R2 in our analysis: it is the only source whose median per-pixel polarization-angle offset exceeds the adopted 10° tolerance (signed offset $+8.8^\circ$), and it also exhibits a strongly negative spectral-index difference $\gamma - \alpha = -0.96$ (§3.3)—departing in the same direction as the survey-wide trend rather than reversing it (the most extreme negative value in the spectral-index sample is L1688 Ophiuchus A at $\gamma - \alpha = -1.27$). Mon R2 was independently excluded from the SOFIA/HAWC+ polarization-angle comparison owing to large measurement uncertainties (Paper T §3.4). In units of the per-pixel statistical detection limit (median $\sigma_{\Delta\chi} \approx 1.2^\circ$ for Monoceros R2), the median $|\Delta\chi|$ corresponds to a characteristic $\sim 9\sigma$ per-pixel deviation, far above the survey-wide $3\sigma_{\Delta\chi} \approx 3.4^\circ$ resolution. We therefore treat Monoceros R2 as the strongest single-source stress test on the wavelength-independent field geometry assumption, and revisit its physical interpretation in §4.2 in the context of size-dependent grain alignment.

Implication.—The PA agreement across 12 of 13 targets supports the interpretation that the 450 and $850 \mu\text{m}$ polarization probes sample the same plane-of-sky magnetic-field structure along each line of sight. Since submillimeter emission at both wavelengths is weighted by dust temperature along the column, the parallel polarization angles further indicate that the aligned grain populations probed at the two wavelengths occupy the same physical regions. Any wavelength dependence of the polarization fraction or of the polarized-intensity spectral index documented in the subsections that follow can therefore be attributed to genuine grain-physics effects rather than to geometric artefacts from probing different field structures at different wavelengths. Similar wavelength-independent magnetic-field morphology has been reported in previous single-dish and interferometric studies of individual star-forming regions (e.g., Vaillancourt & Matthews 2012; Hull et al. 2014). With this precondition established, §3.2 examines the polarization-fraction wavelength dependence directly through the per-pixel ratio $R_p \equiv P_{450}/P_{850}$, and §3.3 disaggregates that ratio into the spectral-index pair (α, γ) .

3.2. Polarization-fraction ratio

We use the polarization-fraction ratio $R_p \equiv P_{450}/P_{850}$ as a column-independent, dual-band observable of differential grain-alignment efficiency: its purpose in this paper is to isolate the intrinsic wavelength-dependent polarization-fraction content of the dust column from the spatial variations in column density and temperature that dominate any single-band P_ν map. Because $P_\nu \equiv PI_\nu/I_\nu$ is already a per-pixel ratio in which the dust column approximately cancels, R_p retains only the wavelength-dependent grain-alignment content: a departure from unity, $R_p \neq 1$, signals

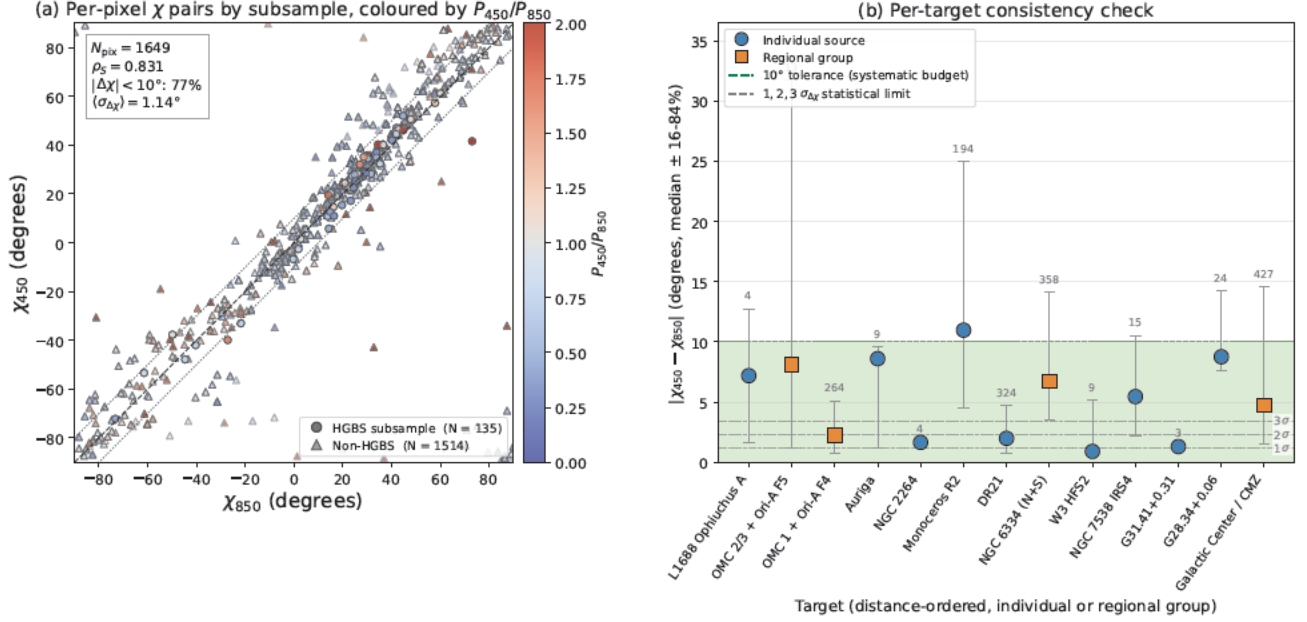


Figure 2. Validation of wavelength-independent magnetic field orientation between 450 and 850 μm . (a) Per-pixel scatter of polarization angles (χ_{850} , χ_{450}), colored by the polarization-fraction ratio $R_p = P_{450}/P_{850}$, for all pixels with valid P_{450}/P_{850} at the matched $12''6$ resolution across the 21 sources with per-pixel data ($N_{\text{samp}} = 1649$ in total). Circles identify pixels drawn from the HGBS 9-field subsample introduced in §3.2, which underlies the per-pixel correlation analyses of Figures 4 and 5; triangles mark the remaining non-HGBS pixels. The close coincidence of the two populations along the 1:1 ridge supports generalizing the HGBS-subsample correlation results to the broader survey. The diverging *coolwarm* colormap is centered at $R_p = 1$ (white), so pixels shifted toward blue (red) carry $R_p < 1$ ($R_p > 1$); coloring by R_p provides an independent third dimension on top of the two axes, which already encode χ_{850} and χ_{450} . The dotted lines mark the $\pm 10^\circ$ physical tolerance band. The corner box reports the global Spearman correlation ρ_s , the fraction of pixels within the tolerance band, and the median per-pixel statistical detection limit ($\sigma_{\Delta\chi}$). (b) Per-target consistency check. For each “target”—an individual source (blue circles) or a regional group of physically associated sources (orange squares)—we show the median absolute angle difference $|\chi_{450} - \chi_{850}|$ together with its 16th–84th percentile envelope. Targets are ordered by representative distance. The light-green band marks the 10° physical tolerance scale, the combined budget of beam matching, offset correction residuals, spatial filtering, and SOFIA cross-validation; see the main text for the breakdown. The three thin gray dashed lines mark the survey-wide statistical detection limit at $1\sigma_{\Delta\chi}$, $2\sigma_{\Delta\chi}$, and $3\sigma_{\Delta\chi}$, where $\sigma_{\Delta\chi} \approx 1.1^\circ$ is the median per-pixel formal uncertainty derived from the Stokes Q , U noise model (companion Paper T §3.3). The small number above each point gives N_{samp} for that target. The grouping combines physically associated fields: the spatially overlapping *OMC 1 + Ori-A F4*; the similar intermediate-mass star-forming pair *OMC 2/3 + Ori-A F5*; the two NGC 6334 sub-fields, denoted *NGC 6334 (N+S)*; and the Central Molecular Zone clouds, denoted *Galactic Center / CMZ*. Twelve of the thirteen targets fall within the 10° tolerance band. However, several targets show small ($< 10^\circ$) but statistically resolved offsets above the $3\sigma_{\Delta\chi}$ line, including L1688 Ophiuchus A, OMC 1 + Ori-A F4, NGC 6334 (N+S), and OMC 2/3 + Ori-A F5. The single target exceeding the 10° tolerance, Monoceros R2, corresponds to a $\sim 9\sigma$ per-pixel deviation and is discussed in the text and in §4.

that the grain populations dominating the polarized emission at 450 and 850 μm polarize with different efficiencies. A qualitatively consistent wavelength-dependent polarization-fraction trend is seen in independent multiwavelength polarimetry of NGC 2071 combining SOFIA/HAWC+ at 154 and 214 μm with JCMT/POL-2 at 850 μm , attributed there to size-dependent grain alignment (Fanciullo et al. 2022); the 450/850 μm R_p analysis presented here extends this continuity across the POL-2 bands and over a much larger sample. The value $R_p < 1$ thus constitutes one of three observational findings (F1–F3) that jointly constrain submillimeter grain-alignment physics; the 450 μm offset-correction and its SOFIA/HAWC+ cross-validation that underpin the input maps are documented in §2 and Paper T §3.4. In what follows we (i) define the canonical Method 3 R_p estimator and the high-quality dual-band mask it inherits, (ii) report the predominant behavior $R_p < 1$ at the survey level, (iii) quote the 28-source sample median, and (iv) introduce the polarization-angle dispersion function $\mathcal{S}$ and the P – $\mathcal{S}$ relation as the geometric-depolarization control on R_p .

Canonical estimator.—We adopt Method 3, the equal-weighted median of pixel-wise polarization ratios over the common

dual-band detection mask (Figure 3), as the canonical estimator for the R_p analysis of Paper S1:

$$R_3 \equiv \text{median}_{i \in \mathcal{M}} \left[\frac{P_{450,i}}{P_{850,i}} \right], \quad (1)$$

where $\mathcal{M}$ is the set of pixels satisfying the common dual-band Stokes- I detection mask, $I_{450}/\sigma_{I,450} > 100$ and $I_{850}/\sigma_{I,850} > 100$, and the median is taken over its members per source. We deliberately impose no selection on the polarized-intensity signal-to-noise ratio, so that intrinsically low-polarization pixels are retained; selecting on PI/σ_{PI} would preferentially remove the weak- P tail of the pixel distribution and therefore bias the estimator. Because each term $P_{450,i}/P_{850,i}$ is formed from the same line of sight, the estimator benefits from a generic *common-mode cancellation*: pixel-to-pixel variations in column density, dust temperature, or beam-scale field tangling enter P_{450} and P_{850} in a correlated way, and their common contribution partially cancels in the per-pixel ratio. R_3 is therefore insensitive to the spatial intensity weighting that dominates intensity-weighted estimators, and the residual signal in R_p isolates intrinsic grain physics rather than line-of-sight geometry. This choice keeps the P_{450}/P_{850} analysis internally consistent with the pixel selection used for the spectral-index analysis of §3.3 and for the per-pixel environmental correlations developed later in this paper. Crucially, the $I/\sigma_I > 100$ threshold combined with the $12''$ grid sampling at the $12''/6$ matched beam keeps each retained pixel statistically independent at the beam scale and high signal-to-noise at both wavelengths; the per-pixel ratio common-mode cancellation therefore robustly suppresses the $450 \mu\text{m}$ offset-correction systematics inherited from companion Paper T §3.4, leaving the systematic floor $\sigma_{\text{sys}}(R_3) \lesssim 0.02$ of Paper T §4.2 (summarized in Paper T Table 6) an order of magnitude below the detected signal. The alternative intensity-weighted estimator ($\tilde{R}_1$) and pixel-by-pixel weighted regression (R_2)—whose full derivation and source-by-source tables are presented in companion Paper S2 (§3.3)—yield sample medians consistent with Method 3 within the quoted precision, so restricting Paper S1 to Method 3 does not affect the survey-level conclusions. In practice, Method 3 defines the common high-quality dual-band mask used throughout this paper: it underlies the survey-level R_p estimator (§3.2, Figure 3), the DUALBAND sample (417 pixel pairs of the HGBS-overlap subset; defined in §3.2) used for the per-pixel correlation tests (Figs. 4 and 5), and the SPIDX sample after the additional per-source $N_{\text{samp}} \geq 5$ requirement (§3.3, Figure 6).

Predominant behavior: $P_{450}/P_{850} < 1$.—The full 69-field census presented in companion Paper S2 (§3.1) shows that roughly four-fifths of sources exhibit $R_p < 1$, with a typical value $R_p \approx 0.5$ – 0.7 . This behavior holds across a wide range of environments: nearby low-mass cores (L1688 Ophiuchus A at 139 pc), intermediate-distance high-mass regions (OMC 1 at 399 pc, DR 21 at 1500 pc), and distant Galactic Center clouds (Sagittarius B2 at 8.3 kpc). The consistency of $R_p < 1$ across roughly four-fifths of the sources, spanning nearly two orders of magnitude in heliocentric distance and more than two orders of magnitude in column density, motivates an interpretation in terms of intrinsic grain physics rather than environmental or geometric conditions specific to individual clouds.

Sample median.—Across the SPIDX sample (the 28-source spectral-index subsample; see Table 2 for the F1–F3 sample ascriptions of this paper), the equal-weighted median of pixel ratios yields

$$R_3 = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}, \quad (2)$$

where $\delta R_{\text{sys}} = 0.02$ is the systematic floor of companion Paper T §4.2 and $\delta R_{\text{stat}} = 0.04$ is the empirical standard error of the median estimated from the per-source dispersion of the SPIDX sample. For the highest signal-to-noise source, L1688 Ophiuchus A, $R_3 = 0.47$; several other well-measured sources (e.g., DR 21 with $R_3 = 0.51$, NGC 6334 North with $R_3 = 0.46$, and the Central Molecular Zone (CMZ) stack) lie in the same range. The complete catalog per-source R_3 — and the comparison per-source with $\tilde{R}_1$ and R_2 that confirms the insensitivity of R_p to the choice of weighting scheme — is given in the companion Paper S2 §3.3.

Polarization-angle dispersion function $\mathcal{S}$.—The role of $\mathcal{S}$ in this paper is as the per-pixel control variable for geometric depolarization: a pixel with small $\mathcal{S}$ contains a locally ordered plane-of-sky field, whereas a pixel with large $\mathcal{S}$ contains a tangled field that will beam-average to a reduced polarization fraction regardless of grain physics. We therefore use $\mathcal{S}$ to test whether the observed $R_p < 1$ can be explained by geometry alone, through the per-band P – $\mathcal{S}$ relation of §3.2 and the geometry-cancelling proxy $\mathcal{S} \times \mathcal{P}^{\text{frac}}$ above. Operationally, we quantify magnetic-field coherence within the beam and along the line of sight through the polarization-angle dispersion function (Hildebrand et al. 2009; Planck Collaboration 2020):

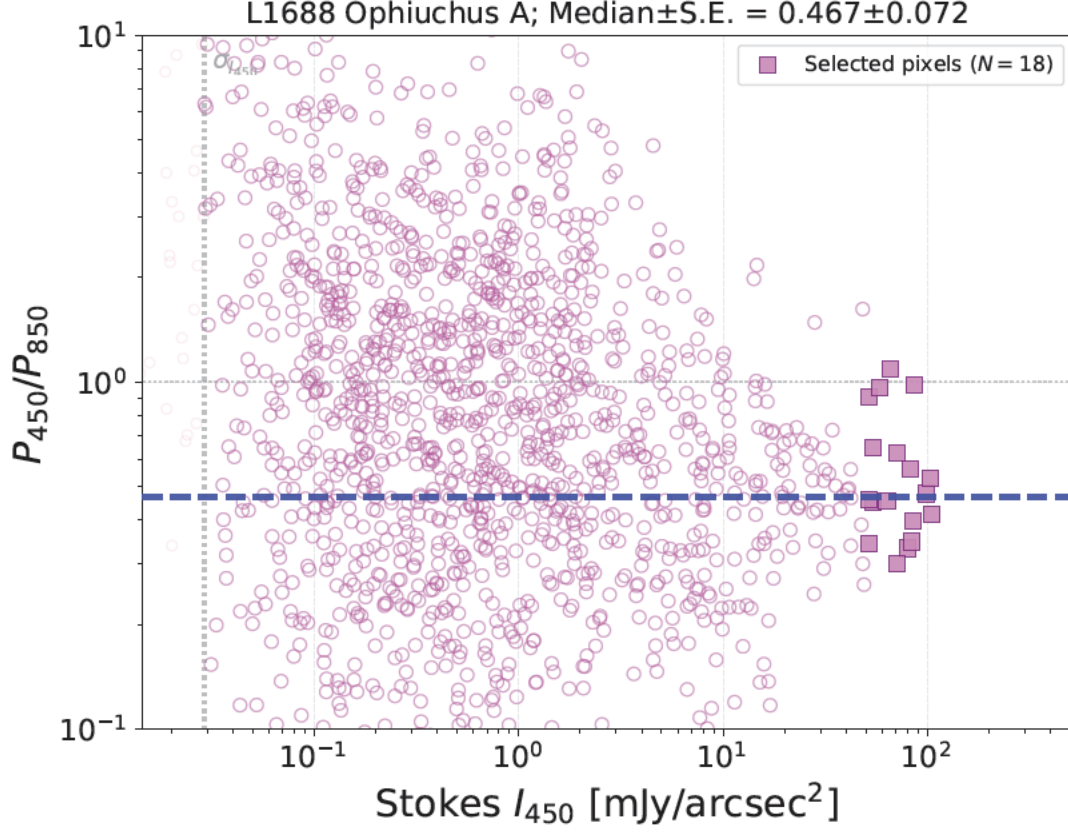


Figure 3. Illustration of the Method 3 selection criterion for L1688 Ophiuchus A. Individual pixel ratios P_{450}/P_{850} are plotted against Stokes I_{450} ; filled violet squares are pixels satisfying the Stokes- I detection mask $I/\sigma_I > 100$ at both wavelengths, with no selection imposed on the polarized-intensity signal-to-noise ratio, open magenta circles are excluded pixels, and the horizontal dashed line marks the Method 3 median $R_3 = 0.47 \pm 0.07$ of the selected pixels; here “S.E.” in the in-figure title corresponds to the statistical error δ_{stat} . For this source, $R_1 = 0.49$, $R_2 = 0.47$, and $R_3 = 0.47$ agree to within 0.02, so the choice of weighting scheme does not affect the inferred R_p . The full per-source comparison is presented in companion Paper S2 (§3.1).

$$\mathcal{S} = \sqrt{\frac{1}{n_{\text{pix}} - 1} \sum_{k=1}^{n_{\text{pix}}} |\phi_k - \phi_{\text{center}}|^2 - \sigma_\phi^2}, \quad (3)$$

where ϕ_k is the polarization angle at the k -th neighbor pixel (wrapped to $(-90^\circ, +90^\circ)$), ϕ_{center} is the center-pixel angle, and σ_ϕ^2 is the noise-bias correction inherited from the Stokes Q, U uncertainty budget of companion Paper T §3.3 (see also Montier et al. 2015; Planck Collaboration 2020). We adopt a pixel-centered 3×3 kernel ($36'' \times 36''$ on the sky) with $n_{\text{pix}} \geq 5$ valid neighbors, and take $\mathcal{S}$ as $|\mathcal{S}|$ throughout in the Hildebrand-type convention; the sample-wide range is $\mathcal{S}_{\text{min}} \approx 0.5^\circ$ (set by measurement noise) to $\mathcal{S}_{\text{max}} \approx 52^\circ\text{--}63^\circ$ depending on the exact form of the definition. $\mathcal{S}$ is computed identically at 450 and $850\,\mu\text{m}$, so differences in $\mathcal{S}$ between bands reflect genuine field-structure differences rather than algorithmic choices.

Grain-alignment efficiency proxy $\mathcal{S} \times \mathcal{P}^{\text{frac}}$.—A complementary diagnostic following Fissel et al. (2016), $\mathcal{S} \times \mathcal{P}^{\text{frac}}$, is constructed so that beam-scale field-tangling cancels in the product and any persistent variation reflects intrinsic grain-alignment efficiency. Because a meaningful interpretation of this proxy requires dynamic range in both $\mathcal{S}$ and $\mathcal{P}^{\text{frac}}$ that only the full 69-source census provides—including environments where $\mathcal{S}$ spans its sample-wide range—its per-source distribution is developed in companion Paper S2 §3.2 and its environmentally stratified analysis in Paper S2 §3.5. In the present paper the proxy is referenced only as interpretive framing for the geometric-depolarization control of §3.2.

Geometric-depolarization control via the P - $\mathcal{S}$ relation.—To test whether the observed $R_p < 1$ is driven by geometric depolarization—beam-averaging of tangled magnetic fields—we analyze the correlation between the polarization fraction P and the polarization-angle dispersion function $\mathcal{S}$ at each wavelength.

For this purpose, we define a matched-resolution per-pixel subsample drawn from the nine of our survey regions that happen to fall within the footprint of the *Herschel* Gould Belt Survey (HGBS; André et al. 2010): L1688 Ophiuchus A, Taurus B211/213, NGC 1333, Perseus B1, Perseus IC348, OMC 1, OMC 2, OMC 3, and Serpens Main. We emphasise that the identity of these nine regions is set by *availability* of publicly released HGBS N_{H_2} and T_{d} maps at the positions of our POL-2 fields, not by any physics-driven selection applied to our sample; the HGBS-matched subsample therefore inherits the sample-selection function of the parent 69-field survey (§2) without additional bias toward a particular environmental regime. This HGBS-matched subsample provides co-spatial maps of column density N_{H_2} and dust temperature T_{d} at angular resolutions comparable to the POL-2 data, spanning $\sim 10^{22}\text{--}10^{24}\text{ cm}^{-2}$ and $\sim 10\text{--}45\text{ K}$ across the pooled pixel set.

We restrict the analysis to Method 3 pixels within these regions, yielding a common set of 417 pixels (the DUALBAND sample, in the short-name convention of Paper T Table 2; equivalently *417 independent measurements*, since the $12''$ grid is comparable to the matched $12''6$ beam) used consistently in this section and in the spectral-index analysis of §3.3. Both bands are pre-matched to a common $12''6$ resolution (§2), so the $12''$ grid samples identical effective beams at both wavelengths.

Applying the per-pixel detection mask of companion Paper T §3.3 ($P_i \geq 3\delta P_i$ in each band, with $\mathcal{S}$ taken as $|S|$ in the Hildebrand-type convention), modest but statistically significant negative ranked correlations are detected at both wavelengths (Figure 4),

$$r_s(P_{450}, \mathcal{S}_{450}) = -0.19 \quad (N = 345, p = 5.0 \times 10^{-4}), \quad (4)$$

$$r_s(P_{850}, \mathcal{S}_{850}) = -0.14 \quad (N = 372, p = 5.7 \times 10^{-3}), \quad (5)$$

where both subsamples are drawn from the DUALBAND sample (417 independent pixel-pair measurements of the HGBS-overlap subset) after applying the per-band detection mask; the count at $850\text{ }\mu\text{m}$ further excludes a small number of pixels with undefined angle dispersion at that wavelength. Removing OMC 1—the warmest and most luminous single source in the HGBS subsample—from the above-floor sample collapses both correlations to statistical insignificance: $r_s = -0.05$ ($N = 283, p = 0.38$) at $450\text{ }\mu\text{m}$ and $r_s = -0.03$ ($N = 310, p = 0.58$) at $850\text{ }\mu\text{m}$. The modest sample-wide correlation of Equations (4)–(5)— $|r_s| \approx 0.15\text{--}0.2$ —is therefore driven primarily by the high-radiation OMC 1 regime (characterized in our HGBS subsample by $T_{\text{d}} \gtrsim 30\text{--}40\text{ K}$ and the highest N_{H_2} bin); outside that regime the per-band relation $P\text{--}\mathcal{S}$ is consistent with statistical noise. Within the OMC 1 regime, the existence of negative ranked correlations at *both* wavelengths establishes that geometric depolarization is operative there, and because the two bands are evaluated on the same pixels, their common contribution partially cancels in the ratio R_{p} . The $P\text{--}\mathcal{S}$ relation therefore functions as a *presence test* for geometric depolarization within the OMC 1 high-radiation regime rather than as a survey-wide quantitative measure of its magnitude: within that regime, it confirms that geometric tangling is at work in both bands, supporting the cancellation argument that isolates the intrinsic grain-physics contribution to R_{p} .

Caveat: sample-wide modesty and the OMC 1 dependence.—Equations (4) – (5) already apply the Paper T detection-limit mask on P at each band, so they are not inflated by floor-bounded-pixel scatter; therefore, their modest magnitude ($|r_s| \lesssim 0.2$) together with the OMC 1 dependence demonstrated above jointly justify treating the $P\text{--}\mathcal{S}$ relation as a qualitative presence test for geometric depolarization within the OMC 1 regime rather than as a survey-wide quantitative measure. The same OMC 1-contingent behavior appears in the correlations per-pixel $T_{\text{d}}\text{--}P$ and $N_{\text{H}_2}\text{--}P$ tabulated in Appendix B (Table 5) and discussed in §4.3; the dual-wavelength ratio R_{p} cancels the common OMC 1 contribution to all of these.

3.3. Spectral Indices of Total and Polarized Intensity

The polarization-fraction ratio analysis (§3.2) established that $P_{450}/P_{850} < 1$ is the dominant behavior in our sample. Spectral-index analysis provides an interpretation that is complementary to the R_{p} picture: whereas R_{p} collapses the two wavelengths into a single ratio in which the dust column largely cancels, the spectral indices α and γ quantify how total and polarized intensity scale with frequency *separately*, exposing the two wavelengths' contributions to dust-emissivity models and grain-alignment physics individually. We summarize here the framework and the headline F2 results from the SPIDX sample; the full per-source spectral-index catalog, the bias-assessment diagnostics, the spatially-resolved L1688 Ophiuchus A illustration, and the per-environment LMSF/HMSF/CMZ breakdown are presented in companion Paper S2 (Paper S2 §3.4). In what follows we lay out the framework and its connection to R_{p} via Equation (7), present the per-pixel $I_{450}/I_{850}\text{--}R_{\text{p}}$ anticorrelation (Figure 5) as the per-pixel manifestation of $\gamma - \alpha$,

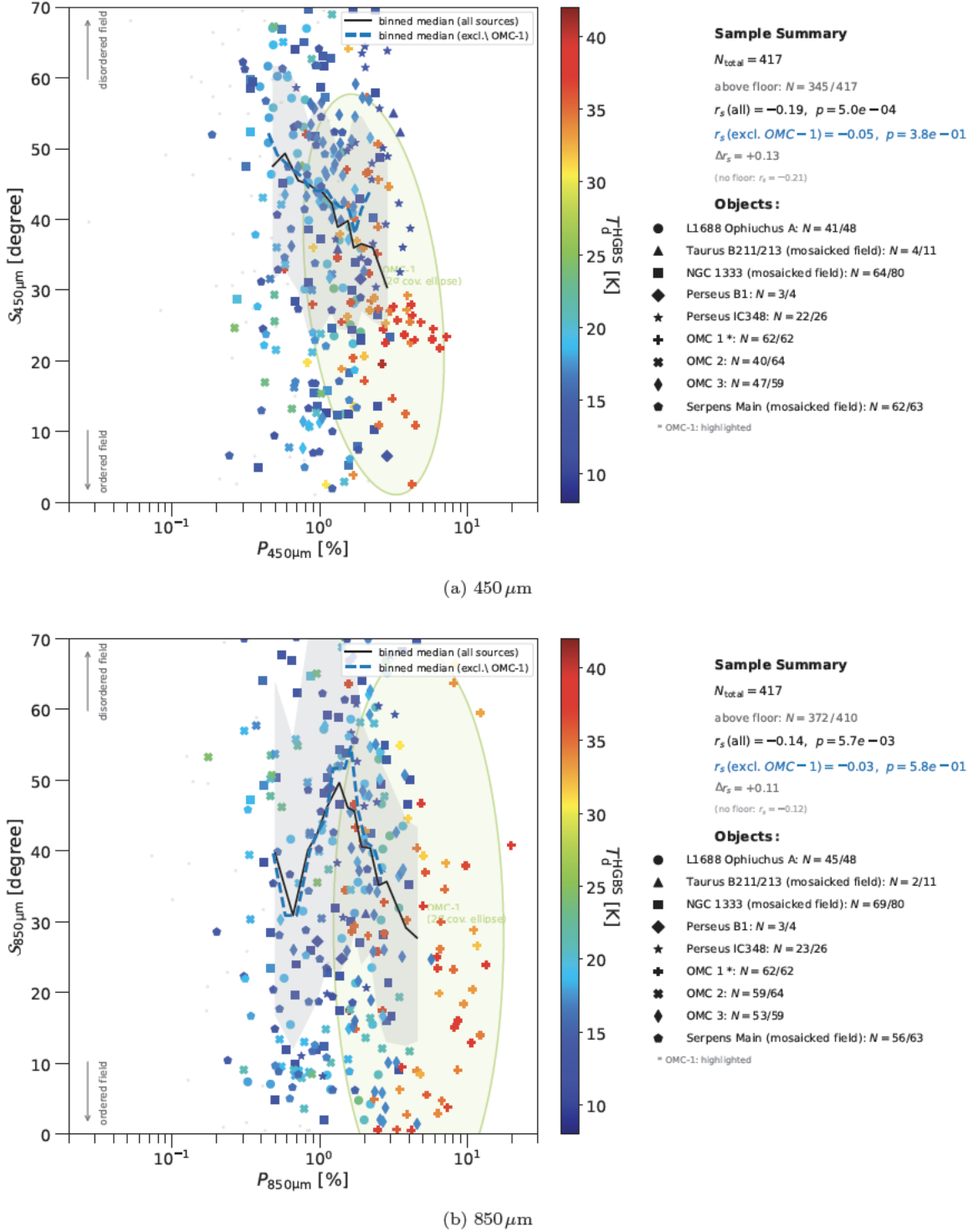


Figure 4. Polarization fraction versus polarization-angle dispersion function $|S|$ at 450 μm (a) and 850 μm (b) for the Method 3 pixels drawn from nine HGBS-covered nearby regions, color-coded by HGBS dust temperature. Applying the per-pixel detection mask of companion Paper T §3.3 ($P_i \geq 3\delta P_i$ at each band), modest but statistically significant negative ranked correlations are detected at both wavelengths ($r_s = -0.19$, $N = 345$, $p = 5.0 \times 10^{-4}$ at 450 μm ; $r_s = -0.14$, $N = 372$, $p = 5.7 \times 10^{-3}$ at 850 μm). Removing OMC 1 from the above-floor sample collapses both correlations to statistical insignificance (§3.2), indicating that the modest sample-wide correlation— $|r_s| \approx 0.15$ – 0.2 above the floor—is driven primarily by the OMC 1 high-radiation regime ($T_d \gtrsim 30$ – 40 K in this HGBS subsample). Within that regime the common action of geometric depolarization on the same pixels partially cancels in the ratio R_p , so the residual $R_p < 1$ behavior cannot be attributed to geometric depolarization alone. We have verified the above floor-dependence by recomputing $r_s(P_i, S_i)$ across a range of polarized-intensity floor choices (nominal $3\sigma_P$, relaxed $2\sigma_P$, and stricter $4\sigma_P$); the resulting shift in r_s is $|\Delta r_s| \lesssim 0.02$ at both wavelengths, well below the reported $|r_s|$ and negligibly small compared to the OMC-1-driven collapse from ≈ -0.19 to ≈ -0.05 at 450 μm (and ≈ -0.14 to ≈ -0.03 at 850 μm), so the narrative does not depend on the particular per-pixel floor adopted. Guide arrows on the left edge identify the “ordered field” (low S) and “disordered field” (high S) regimes.

report the F2 headline result $\gamma < \alpha$ in 75% of the SPIDX sample (Figure 6), and close with robustness and S/N -bias diagnostics; the full per-source bias-check atlas is deferred to companion Paper S2 (§3.2 and App. M).

Framework.—We define the spectral indices for total and polarized intensity as

$$\alpha \equiv \frac{\log(I_{450}/I_{850})}{\log(\nu_{450}/\nu_{850})} \quad \text{and} \quad \gamma \equiv \frac{\log(P_{450}/P_{850})}{\log(\nu_{450}/\nu_{850})}, \quad (6)$$

where $\nu_{450} = 666 \text{ GHz}$ and $\nu_{850} = 353 \text{ GHz}$ are the central frequencies of the two POL-2 bands. Throughout this section we apply the spectral-index analysis to the same Method 3 pixel mask of §3.2 ($I/\sigma_I > 100$ at both wavelengths, no PI cut), which both ensures statistical consistency between the polarization-fraction-ratio and spectral-index per-pixel observables and selects the DUALBAND sample (417 pixels of HGBS overlap) for which independent column-density and dust-temperature constraints are available.

Connection to the polarization-fraction ratio.—Decomposing the polarization-fraction ratio of §3.2 into the spectral indices α and γ links the observational result directly to the grain-physics framework adopted throughout the paper:

$$\frac{P_{450}}{P_{850}} = \left(\frac{\nu_{450}}{\nu_{850}} \right)^{\gamma - \alpha} = 1.89^{\gamma - \alpha}, \quad (7)$$

so that $P_{450}/P_{850} < 1$ requires $\gamma < \alpha$ and a constant P_{450}/P_{850} corresponds to a fixed $\gamma - \alpha$. Equation (7) is cross-checked against the spatially-resolved measurements of α and γ developed in the F2 result paragraph below: the independently measured per-pixel median $\gamma - \alpha = -0.52$ implies $R_p = 0.72$, in good agreement with the directly measured Method 3 sample median $R_3 = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$ of §3.2 within the combined systematic and statistical uncertainties of the two approaches.

For modified-blackbody emission, the total-intensity spectral index can be expressed as $\alpha = \beta + C(T_d)$, where $C(T_d)$ is the Planck correction term. The full derivation and calibration framework are presented in companion Paper T §3.1.

Importantly, the primary observable used in this work, $\gamma - \alpha$, is independent of dust temperature. The principal conclusions of this paper therefore do not depend on any particular assumed dust temperature distribution. Per-source dust temperatures, and any environmentally-stratified β comparisons derived from them, are reported in companion Paper S2 (Paper S2 §3.4). The dust-temperature uncertainty propagates into a per-source β broadening of $\sigma_{\beta,T} \approx 0.35$ (Paper S2 §3.4.1) but cancels exactly in the $\gamma - \alpha$ difference used as the primary observable here.

Per-pixel manifestation: the I_{450}/I_{850} – R_p anticorrelation.—A per-pixel expression of the structure $\gamma - \alpha$ structure implied by Equation (7) is provided by the Spearman rank correlation between the intensity ratio I_{450}/I_{850} and the polarization ratio $R_p = P_{450}/P_{850}$ for the DUALBAND sample (417 independent pixel-pair measurements of the HGBS-overlap subset) introduced in §3.2. A statistically significant negative correlation is detected,

$$r_s(I_{450}/I_{850}, R_p) = -0.49 \quad (N = 329, p = 1.9 \times 10^{-21}), \quad (8)$$

and the anticorrelation remains $r_s = -0.35$ ($N = 267$, $p = 4.4 \times 10^{-9}$) when OMC 1 is excluded, demonstrating that the intensity-ratio–polarization-ratio behavior reflects a general property of the HGBS subsample rather than being driven by a single extreme source. Pixels with higher I_{450}/I_{850} (corresponding to colder dust along the Planck curve between 450 and 850 μm) tend to have R_p values closer to unity, while lower I_{450}/I_{850} pixels (warmer dust) show lower R_p : through Equation (7), this is a direct per-pixel manifestation of the $\gamma - \alpha$ structure, with the same size-dependent alignment physics that produces $\gamma < \alpha$ globally also producing the cold/warm R_p contrast pixel-by-pixel. Clearly, the difference in polarization-angle $|\chi_{450} - \chi_{850}|$ remains small ($\lesssim 20^\circ$) throughout the entire range of I_{450}/I_{850} and R_p , and no correlation is detected between R_p and $|\Delta\chi|$. Geometric depolarization is clearly present at both wavelengths, but its contribution is largely common-mode and therefore partially cancels in the ratio R_p . The absence of any residual correlation with $|\Delta\chi|$ indicates that geometry is not the dominant driver of the observed wavelength dependence, although a subdominant contribution cannot be entirely excluded. The visual presentation of this per-pixel test, showing R_p versus I_{450}/I_{850} with $|\Delta\chi|$ color coding and the per-pixel detection floors superposed, is given in Figure 5 below; the same panel without the floor bands is reproduced in companion Paper S2 (Paper S2 §3.3).

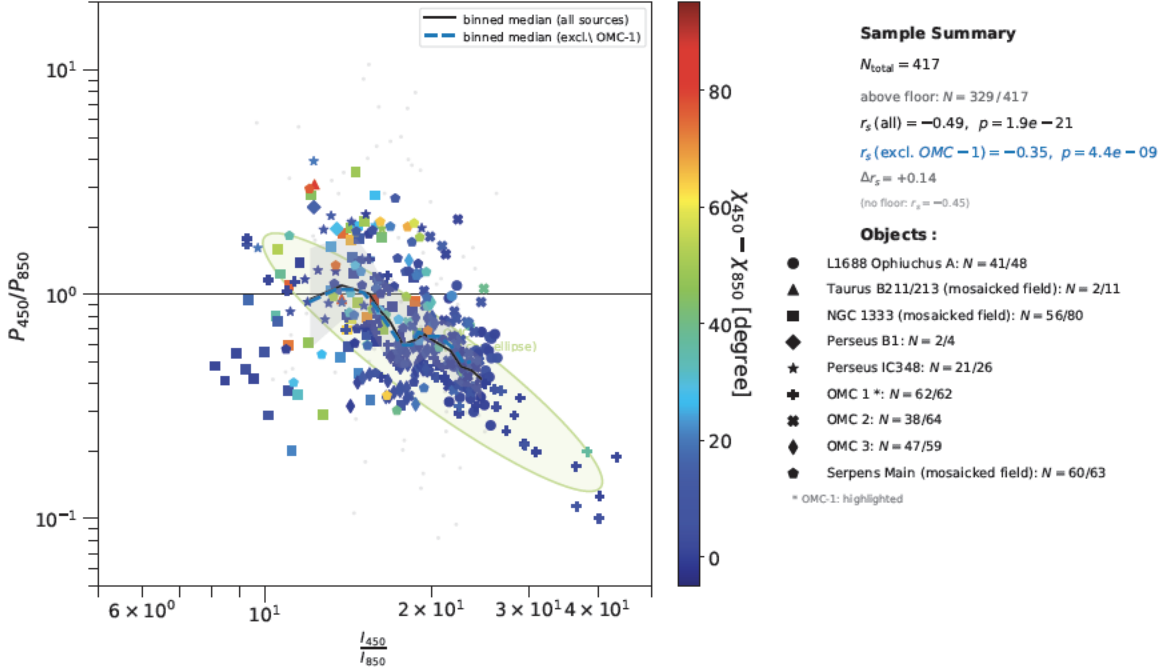


Figure 5. Per-pixel anticorrelation between the intensity ratio I_{450}/I_{850} and the polarization ratio $R_p = P_{450}/P_{850}$ for the DUALBAND sample (417 independent pixel-pair measurements of the HGBS-overlap subset), colored by the polarization-angle difference $|\chi_{450} - \chi_{850}|$. The anticorrelation persists across both the OMC 1-included and OMC 1-excluded subsamples ($r_s = -0.49$, $N = 329$ for the full subsample; $r_s = -0.35$, $N = 267$ excluding OMC 1), demonstrating that the relation reflects a general property of the HGBS subsample rather than being driven by a single luminous source. The angle-difference color coding shows that the relation does not arise from a systematic variation of $|\Delta\chi|$ across the diagram. Shaded vertical and horizontal bands indicate the per-pixel detection limits corresponding to the adopted signal-to-noise threshold ($P_i \geq 3\delta P_i$; companion Paper T §3.3). Through Equation (7), this anticorrelation is a direct per-pixel manifestation of the $\gamma - \alpha$ spectral-index structure developed in the F2 result paragraph below.

F2 result: $\gamma < \alpha$ in 75 % of the statistical sample.—After applying the source-selection criteria of the companion [Paper S2](#) (§3.4 and App. B), the 28 sources with $N_{\text{samp}} \geq 5$ form the statistical sample on which the F2 analysis is based.

Across this statistical sample, the per-source median Stokes I index and the polarized-intensity index are

$$\langle \alpha \rangle = 3.46 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}} \quad (\text{range } 2.71\text{--}4.63), \quad (9)$$

$$\langle \gamma \rangle = 3.12 \pm 0.04_{\text{sys}} \pm 0.11_{\text{stat}} \quad (\text{range } 2.48\text{--}4.42), \quad (10)$$

respectively, across the 28 sources; these per-source medians are the primary observables used throughout this paper, with δ_{sys} taken from the companion Paper T §4.1 budget on $\sigma(\gamma - \alpha) \lesssim 0.04$ (the individual α and γ floors are at most as large as the joint floor, since the offset-correction systematics are common-mode at first order) and δ_{stat} adopted as an empirical standard error of the per-source median, estimated from the per-source dispersion divided by $\sqrt{28}$ (“empirical SEM”). The per-pixel median spectral-index difference is

$$\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}} \quad (\text{range } -1.44 \text{ to } +0.87), \quad (11)$$

where $\delta_{\text{stat}} = 0.10$ is the empirical SEM derived from the per-source dispersion of $\gamma - \alpha$ values. Via Equation (7) this corresponds to $P_{450}/P_{850} = 1.89^{-0.52} = 0.72 \pm 0.02_{\text{sys}} \pm 0.05_{\text{stat}}$, recovering the cross-check against the directly-measured R_3 of §3.2 introduced in the Connection paragraph above. The F2 flagship result is that 21 of 28 sources (75 %) lie below the $\gamma = \alpha$ line, i.e. $\gamma < \alpha$, as visualized in Figure 6.

A small minority of candidates (six of 28) sit above the $\gamma = \alpha$ line in Figure 6 ($\gamma > \alpha$, corresponding to $P_{450}/P_{850} > 1$); however, once the combined systematic floor $\sigma(\gamma - \alpha) \lesssim 0.04$ of companion Paper T §4.2 and the per-source statistical error ($\delta_{\text{stat}} \sim 0.1$, dominated by the γ measurement) are propagated, the majority of these candidates are consistent with $\gamma \leq \alpha$ at the 1σ level, with only the Auriga field exhibiting per-source error bars that do not cross $\gamma = \alpha$.

Per-source identification, environmental context, and the CMZ-specific $\gamma > \alpha$ pattern are developed in companion Paper S2 (Paper S2 §3.4 and §3.5), with the physical interpretation in the context of size-dependent grain alignment given in §4.

Robustness.—The systematic-uncertainty floors of companion Paper T §4.2 (Paper T Table 6), $\sigma(P_{450}/P_{850}) \lesssim 0.02$ and $\sigma(\gamma - \alpha) \lesssim 0.04$, are the δ_{sys} values adopted for Equation (11) and the propagated P_{450}/P_{850} above; both are dominated by 450 μm offset-correction residuals. These floors are an order of magnitude below the detected signals $P_{450}/P_{850} \approx 0.4\text{--}1.4$ and $|\gamma - \alpha| \approx 0.52$, so the flagship behavior of F2 is *resolved at the $\sim 13\sigma_{\text{sys}}$ level* against the systematic floor and is simultaneously resolved against the empirical per-source statistical scatter ($\delta_{\text{stat}} \approx 0.10$). In particular, neither the sign of the effect ($\gamma < \alpha$ in 75% of sources) nor its central magnitude ($\gamma - \alpha \approx -0.52$) can be inverted or eliminated by any plausible revision of the 450 μm offset-correction, because no such revision would be comparable to the signal. The consistency between the spectral-index method and the per-method polarization-fraction ratios of §3.2 at the per-source level, demonstrating mutual agreement within combined systematic and statistical uncertainties for L1688 Ophiuchus A and other well-measured sources, is detailed in companion Paper S2 (Paper S2 §3.4).

S/N-dependent bias assessment.—Because polarized-intensity measurements are intrinsically weaker than total intensity, the polarized-intensity spectral index γ is more susceptible to S/N -dependent biases than the Stokes I spectral index α : at low S/N , the modified-asymptotic debiasing applied to PI does not completely remove the residual Ricean bias, so fluctuations in PI_{450} and PI_{850} can propagate as a systematic downward bias in γ . To mitigate this, we perform a source-by-source running-median analysis of γ (and α) against $S/N(\gamma)$ (resp. $S/N(\alpha)$) using logarithmic S/N bins ($\Delta \log_{10}(S/N) = 0.2$), and adopt per-source $S/N(\gamma)$ thresholds in the range 5–7 above which the running median becomes statistically consistent with a constant value within its formal uncertainties. Representative examples—high- S/N DR 21, the extreme- $\gamma - \alpha$ Monoceros R2, the $\gamma > \alpha$ Serpens Main, the low- N_{samp} OMC 3, and the JCMT calibrator G31.41+0.31—all yield running medians that stabilise above $S/N \gtrsim 5\text{--}7$, confirming that the adopted thresholds effectively mitigate S/N -dependent biases across diverse data-quality and spectral-index regimes. Full per-source bias-check diagnostics, including an L1688 Ophiuchus A reference walk-through and a bias-check atlas covering the full SPIDX sample, are presented in companion Paper S2 (§3.4 and App. B).

Taken together, the three flagship findings of §3.1–§3.3 — the pixel-level wavelength-independence of the magnetic-field orientation across 12 of 13 targets (F1, §3.1), the universal $R_p < 1$ with its geometric-depolarization control (F3, §3.2), and the systematic $\gamma < \alpha$ observed in 75% of the statistical sample (F2, §3.3) — establish that the aligned grain populations probed at 450 and 850 μm occupy the same physical regions yet exhibit a shallower effective polarized-intensity spectrum than the bulk dust. The most parsimonious interpretation of this combined evidence is a wavelength-dependent alignment efficiency linked to the grain-size distribution, although alternative explanations such as compositional variations or multi-component temperature structure cannot be entirely excluded. The full physical development, including a discussion of the high-radiation regime that bounds our sample at the warm end, is presented in §4.

4. DISCUSSION

Our dual-wavelength polarimetric survey establishes three observational constraints on submillimeter polarization physics (§4.1): (1) wavelength-independent field orientation, (2) systematic $\gamma < \alpha$, and (3) $P_{450}/P_{850} < 1$ independent of geometric effects. These findings, interpreted in §4.2–§4.3, are most consistent with a picture of size-dependent grain alignment.

4.1. Summary of the Three Findings

The Results and Analysis section (§3) establishes three observational facts that constrain the physical origin of submillimeter polarization. We summarize these findings here in the order in which they enter the physical argument of §4.2–§4.3 rather than in the order they appeared in §3: Finding 1 (B-field angle consistency, from §3.1) establishes the geometric precondition; Finding 2 ($\gamma < \alpha$, from §3.3) provides the principal spectral-index diagnostic; and Finding 3 ($R_p < 1$ independent of geometry, from §3.2) closes the loop by showing that the ratio diagnostic cannot be absorbed by residual geometric effects. This order therefore reflects the progression *geometric control*, *spectral-index evidence*, then *ratio-level confirmation*, not the presentation order of §3.

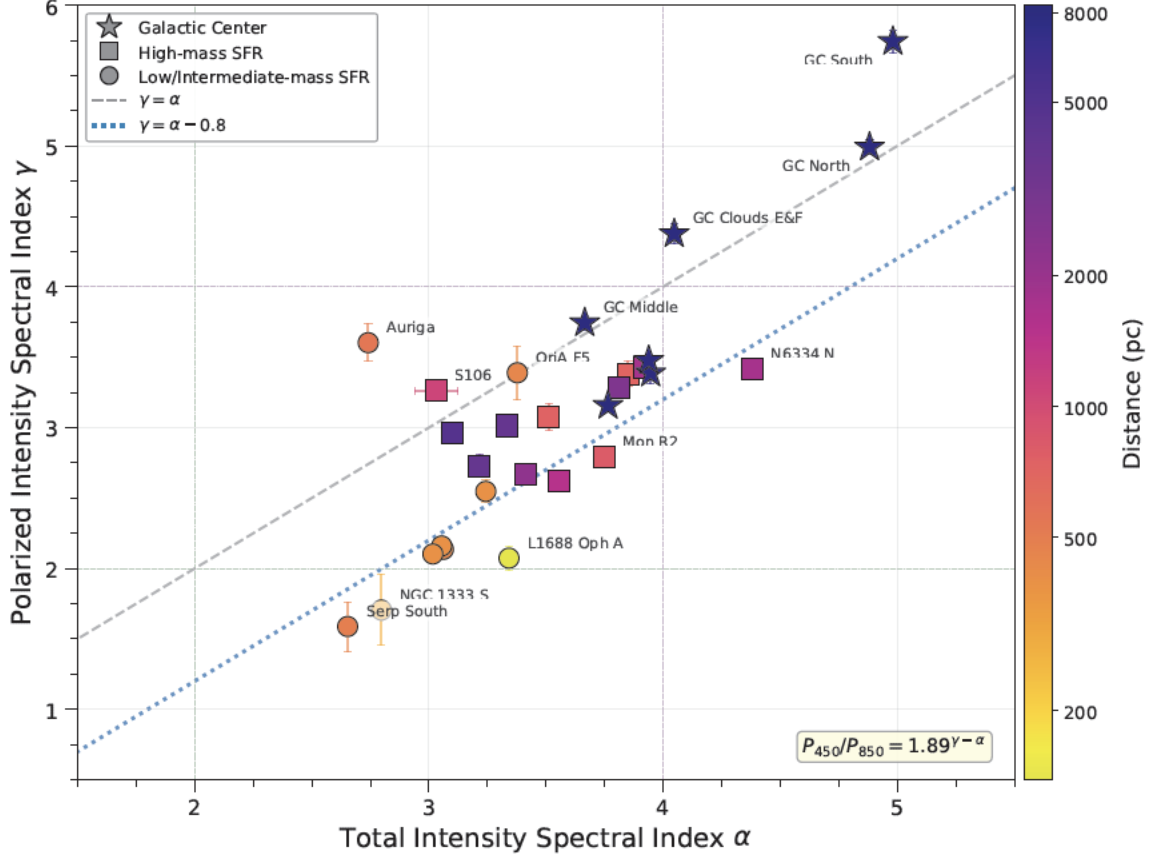


Figure 6. Correlation between the Stokes I spectral index $\langle\alpha\rangle$ and the polarized-intensity spectral index $\langle\gamma\rangle$ for the 28 statistical-sample sources ($N_{\text{samp}} \geq 5$). Twenty-one of 28 sources (75%) lie below the $\gamma = \alpha$ line, with median $\gamma - \alpha = -0.52$ ($P_{450}/P_{850} = 0.72$); per-source medians are $\langle\alpha\rangle = 3.46$ and $\langle\gamma\rangle = 3.12$. The dashed line shows $\gamma = \alpha$; the dotted line shows $\gamma = \alpha - 0.8$ ($P_{450}/P_{850} \approx 0.6$); error bars represent the standard error of the median. Symbols distinguish environment categories (stars: CMZ; squares: high-mass; circles: low/intermediate-mass); the per-environment breakdown and the six $\gamma > \alpha$ sources are presented in companion Paper S2 (Paper S2 §3.4 and §3.5).

Finding 1: Wavelength-Independent Magnetic Field Orientation—The polarization angles at 450 and 850 μm agree to within $|\chi_{450} - \chi_{850}| \lesssim 10^\circ$ across the sample (Figure 2), a value small compared to the 180° angular range and within the combined systematic budget. In the unified notation, a typical pixel of the sample is described by $\Delta\chi = \bar{\Delta}\chi \pm 10^\circ_{\text{sys}} \pm 1.1^\circ_{\text{stat}}$ with $\bar{\Delta}\chi \lesssim 1^\circ$, where the systematic component is the physical tolerance budget and the statistical component is the Stokes Q, U noise-model output (§3.1; companion Paper T §§3.3–3.4). The per-pixel statistical detection limit $\sigma_{\Delta\chi} \approx 1.1^\circ$ is much smaller than the 10° tolerance, resolving small but significant offsets in a few targets, with Monoceros R2 the most extreme case ($\sim 9\sigma$ per-pixel deviation; §3.1). This agreement holds across the full range of intensity ratios, polarization ratios, and environmental conditions in our sample. The most straightforward interpretation is that both wavelengths predominantly trace the same plane-of-sky magnetic field structure, and that the observed wavelength dependence in polarization fraction most likely arises from intrinsic grain properties rather than from probing geometrically distinct field configurations, although localized exceptions may exist in individual regions (e.g., L1688; §4.2).

Finding 2: Systematic $\gamma < \alpha$ Across Diverse Environments—The polarized-intensity spectral index γ is smaller than the total-intensity spectral index α in 21 of 28 sources (75%) with sufficient signal-to-noise for spectral-index determination (Figure 6). Per-source medians are $\langle\alpha\rangle = 3.46 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ and $\langle\gamma\rangle = 3.12 \pm 0.04_{\text{sys}} \pm 0.11_{\text{stat}}$ (§3.3; empirical SEM from per-source dispersion), yielding the per-pixel median

$$\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}.$$

This systematic difference is observed across low-mass cores, intermediate-mass regions, high-mass star-forming regions, and Galactic Center clouds. The implication is that polarized emission has a shallower spectral dependence than the total intensity, consistent with a dominant contribution from the aligned grain population.

Finding 3: $P_{450}/P_{850} < 1$ Independent of Geometric Effects—The polarization-fraction ratio P_{450}/P_{850} is less than unity for the majority of pixels and sources, with the sample median $P_{450}/P_{850} = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$ and per-source values typically spanning the 0.4–1.4 range (§3.2). Critically, this ratio shows no systematic dependence on the polarization-angle difference $|\chi_{450} - \chi_{850}|$ (Paper S2 §3.3), and the per-band P – S Spearman analysis of §3.2 shows that geometric depolarization — operative within the OMC 1 high-radiation regime — largely cancels in R_p through its common-mode behavior across the two wavelengths. If the wavelength-dependent polarization arose from line-of-sight integration through regions with different field orientations, or from beam-averaging of spatially distinct structures, systematic variations in $\Delta\chi$ with P_{450}/P_{850} would be expected. The absence of such variations, combined with Finding 1, strongly suggests that $P_{450}/P_{850} < 1$ reflects intrinsic grain physics—most naturally interpreted as size-dependent alignment efficiency—rather than geometric artefacts.

Logical Structure—These three findings combine to yield a coherent physical picture.

Finding 1 establishes that geometric effects from line-of-sight integration through distinct field structures do not produce the observed wavelength dependence.

Finding 3 confirms that the polarization ratio P_{450}/P_{850} is independent of geometric artefacts, isolating intrinsic grain physics as the origin.

Finding 2 is most parsimoniously interpreted as size-dependent alignment: the aligned grain population appears to exhibit a shallower effective spectral dependence than the bulk dust, producing $\gamma < \alpha$.

In the following subsections, we interpret these findings in terms of grain alignment physics (§4.2) and implications for RAT theory (§4.3), and discuss the discriminating power and remaining degeneracies of our observations (§4.4); the principal results and benchmarks are summarized in §5.

4.2. Size-Dependent Grain Alignment: Origin of $\gamma < \alpha$

The systematic finding that $\gamma < \alpha$ in 75% of sources (21/28; Finding 2; Figure 6) demonstrates that polarized emission has a shallower spectral dependence than total-intensity emission. In this subsection, we argue that this result is most parsimoniously explained by size-dependent grain alignment, where larger grains align more efficiently than smaller grains. We begin by ruling out geometric origins and develop the size-dependent grain-alignment interpretation; remaining dust-physical degeneracies are organized in §4.4.

Ruling Out Geometric Origins—As established in §4.1, Findings 1 and 3 rule out geometric depolarization as the origin of $\gamma < \alpha$: wavelength-independent polarization angles exclude line-of-sight field mixing, while the absence of any R_p – $|\Delta\chi|$ correlation excludes beam-averaging artefacts.

Additionally, geometric depolarization from magnetic field tangling is directly measurable in our data: applying the per-pixel detection mask of Paper T §3.3, polarization fraction shows a modest but statistically significant negative ranked correlation with $|S|$ at both wavelengths ($r_s = -0.19$, $N = 345$, $p = 5.0 \times 10^{-4}$ at 450 μm ; $r_s = -0.14$, $N = 372$, $p = 5.7 \times 10^{-3}$ at 850 μm ; Figure 4), driven primarily by the OMC 1 high-radiation regime and collapsing to statistical insignificance outside OMC 1 (§3.2). Within the OMC 1 regime, this common-mode geometric effect operates at both wavelengths, so the residual $P_{450}/P_{850} < 1$ there cannot arise solely from differential geometric depolarization.

Thus, Finding 2 reflects intrinsic grain properties, not geometric artefacts.

Among the three diagnostics introduced in §1, the P – S negative ranked correlation plays a distinct role: unlike P_{450}/P_{850} and $\gamma - \alpha$, which quantify the wavelength dependence directly and whose signatures persist across both the OMC 1-included and OMC 1-excluded subsamples (§4.3), the P – S relation serves as a *presence test* that demonstrates geometric depolarization is operative at both bands within the OMC 1 high-radiation regime, where its common-mode cancellation in R_p is most directly verifiable.

Sample Selection and the F2 Incidence—The incidence of $\gamma < \alpha$ established in §3.3 should also be interpreted in the context of the survey selection function described in §2. Because reliable 450 μm polarized measurements preferentially select bright, actively star-forming cloud interiors with substantial local radiation fields, the present statistical sample is weighted toward environments where RAT-driven size-selective alignment is expected to operate efficiently. The

predominance of $\gamma < \alpha$ in this sample (75%; 21/28) therefore demonstrates that the effect is common within the high-surface-brightness regime presently accessible to dual-band JCMT polarimetry. Whether the same incidence remains equally strong in colder and more diffuse cloud envelopes, where the shorter-wavelength polarized emission becomes much weaker, is an open question that will require substantially deeper observations.

Monoceros R2 as a Joint Stress Test—Monoceros R2 is the only source whose median polarization-angle difference exceeds the adopted 10° physical tolerance ($|\chi_{450} - \chi_{850}| = 11^\circ$, signed offset $+8.8^\circ$, $\sim 9\sigma$ at the per-pixel statistical detection limit; §3.1), and it also sits well below the $\gamma = \alpha$ line with $\gamma - \alpha = -0.96$ —strongly negative but not the most extreme value in the sample (the most extreme negative $\gamma - \alpha$ is L1688 Ophiuchus A at -1.27). Critically, both deviations point in the same direction as the survey-wide trends rather than reversing them: Mon R2 amplifies F1 and F2 simultaneously rather than offering a counter-example. We therefore treat Monoceros R2 as a joint stress-test source: for the common projected-field-geometry assumption (F1) and for the lower edge of the F2 ($\gamma < \alpha$) distribution. Per-source diagnosis (radiation anisotropy vs. grain growth vs. line-of-sight thermal stratification) requires modelling beyond the scope of the present survey paper. The population-level treatment of Mon R2 alongside the other influential sources (OMC 1 and Sgr B2) in the BISTRO POLSRC census is summarised in companion [Paper S2 Table 11](#).

Physical Origin: Size-Dependent Alignment Efficiency—With geometric effects excluded, the observation $\gamma < \alpha$ implies that the polarized component has a flatter spectral index than the total intensity. This can arise through two complementary mechanisms in a multicomponent dust model.

First, if polarization is dominated by well-aligned large grains (Lazarian & Hoang 2007a; Hoang & Lazarian 2016; Andersson et al. 2015), and if those grains exhibit a shallower effective emissivity spectrum than the smaller grains that dominate the bulk emission, then the polarization-weighted spectral index γ will naturally be smaller than the mass-weighted α . Laboratory measurements and theoretical models of grain coagulation and mantle growth indicate that the far-infrared opacity spectral slope flattens from ~ 2.0 for bare silicates to ~ 1.2 – 1.6 for composite aggregates (Köhler et al. 2015; Ysard et al. 2019), with the flattening driven by the redistribution of the grain-size distribution toward larger composite-aggregate sizes and by the change in effective dielectric response produced by the carbon mantle on the silicate core. If polarized emission preferentially traces such processed, large grains while total intensity includes contributions from a broader size distribution, then $\gamma < \alpha$ follows naturally.

Second, the intrinsic polarization efficiency may decrease with frequency. In the RAT framework, larger grains are expected to align more efficiently in stronger radiation fields (Lazarian & Hoang 2007a; Hoang & Lazarian 2016): large grains contribute more at $850\ \mu\text{m}$ while small grains contribute relatively more at $450\ \mu\text{m}$. This size-dependent alignment produces an effective wavelength dependence of polarization efficiency, reducing γ relative to α even without requiring the same spectral behavior for the aligned and bulk-emitting dust components.

Comparison with Previous Work—The interpretation developed here builds upon the multiwavelength polarimetric study of NGC 2071 by Fanciullo et al. (2022), who combined HAWC+ ($154, 214\ \mu\text{m}$) and POL-2 ($850\ \mu\text{m}$) observations. That study proposed that the observed wavelength dependence of polarization fraction reflects size-dependent grain alignment, with larger grains at longer wavelengths showing higher polarization efficiency.

Our survey confirms and extends this interpretation. The systematic finding that $\gamma < \alpha$ in 75% of sources (21/28) demonstrates that the wavelength-dependent polarization behavior identified in NGC 2071 is a general characteristic of star-forming regions rather than a peculiarity of that particular source. Despite the $12''6$ beam subtending physical scales that vary by nearly two orders of magnitude—from $\sim 0.008\ \text{pc}$ at Taurus-like distances ($139\ \text{pc}$; L1688 Ophiuchus A) resolving dense core substructure, to $\sim 0.5\ \text{pc}$ at the CMZ ($8.3\ \text{kpc}$; Sagittarius B2) averaging over whole clump-scale regions—the similar $R_p < 1$ and $\gamma < \alpha$ behavior across these environments indicates that size-dependent alignment is a widespread, though not necessarily universal, physical characteristic of interstellar dust in star-forming regions.

Environmental Variations—A minority of sources exhibit $\gamma > \alpha$, including several objects in the Galactic Center environment. These exceptions do not alter the dominant survey-wide trend that $\gamma < \alpha$ in the statistical sample, but they indicate that the polarization spectrum is not strictly universal. A detailed per-environment interpretation of these outliers is deferred to companion [Paper S2 \(§ 3.2 and § 3.5\)](#).

Summary—The systematic result $\gamma < \alpha$ across 75% of sources (21/28), combined with the exclusion of geometric effects (Findings 1, 3, and the P – S relation as a presence test for geometric depolarization), establishes that submillimeter polarization is dominated by well-aligned large grains whose polarized emission exhibits a shallower effective spectral

dependence than the bulk dust emission. This size-dependent alignment provides a unified explanation for $P_{450}/P_{850} < 1$ and offers a quantitative benchmark for theoretical dust models.

4.3. Implications for Radiative Alignment Torque Theory

The discussion below uses the RAT framework as a physically motivated interpretation of the observed dual-band discrepancy, rather than as a unique inversion of the grain-size distribution or alignment-efficiency function; remaining dust-property and line-of-sight degeneracies are addressed in §4.4.

RAT theory provides the leading theoretical framework for understanding grain alignment in the interstellar medium (Lazarian & Hoang 2007a; Hoang & Lazarian 2016; Andersson et al. 2015). Our observations are interpreted within the RAT framework: several aspects align with standard predictions, while others highlight interpretive caveats discussed in the following subsections.

Consistency with RAT Predictions—Several aspects of our results align with RAT expectations.

First, the systematic finding that $\gamma < \alpha$ (Finding 2) is consistent with size-dependent alignment efficiency, a core prediction of RAT theory. Hoang & Lazarian (2016) showed that alignment efficiency increases strongly with grain size because larger grains experience greater radiative torques relative to gas damping. Our observation that polarized emission has a lower spectral index than total intensity directly supports this prediction.

Second, the wavelength-independent magnetic field orientation (Finding 1) confirms that both wavelengths trace the same aligned grain population, as expected if alignment is dominated by a single mechanism operating across grain sizes.

Third, the per-pixel anticorrelation between the intensity ratio and the polarization ratio reported in §3.3 ($r_s(I_{450}/I_{850}, R_p) = -0.49$ for the full HGBS subsample, -0.35 excluding OMC 1; Figure 5) provides a complementary projection of the same picture: the same grains that produce $\gamma < \alpha$ globally also produce a shallower P_{450}/P_{850} where the dust is colder pixel-by-pixel, and this behavior persists across both subsamples. The persistence of this dual-wavelength-ratio relation across both OMC 1-included and OMC 1-excluded subsamples, in contrast to the OMC 1-contingent behavior of the per-band T_d - P and N_{H_2} - P correlations (see the Environmental Regimes paragraph below), strengthens the interpretation that the size-dependent grain-alignment signal is a property of the bulk sample rather than of any single luminous source.

Environmental Regimes and the Role of OMC 1—OMC 1—the warmest and most luminous single source in the HGBS subsample—defines the high-radiation boundary of the physical parameter space sampled by our DUALBAND sample of 417 pixel-pair measurements within the HGBS-overlap subset. The per-band T_d - P and N_{H_2} - P correlations — (historically the most-cited diagnostics of grain alignment in submillimeter polarimetric studies; see Andersson et al. 2015; Pattle et al. 2023) — are themselves sensitive to this boundary: the T_d - P_{450} slope flips sign from $r_s = +0.09$ (full HGBS subsample, $N = 348$) to $r_s = -0.32$ ($N = 286$) when OMC 1 is removed; the strong T_d - P_{850} correlation ($r_s = +0.36$, $N = 383$) collapses to $r_s = +0.03$ ($N = 321$); and the N_{H_2} - P anticorrelations likewise strengthen at 450 μm ($r_s = -0.11$ to -0.49) or sign-flip at 850 μm ($r_s = +0.21$ to -0.13) once OMC 1 is excluded (full tabulation in Appendix B, Table 5). Comparison with prior studies in this area is deferred to the environmental paper (companion Paper S2). This is consistent with the standard RAT picture in which strong-radiation regimes such as OMC 1 promote efficient alignment of even relatively small grains, so that the per-band T_d - P and N_{H_2} - P relations acquire OMC 1-specific contributions that are absent in colder, lower-radiation HGBS sources. The dual-wavelength ratio observables ($\gamma - \alpha$, P_{450}/P_{850} , and $r_s(I_{450}/I_{850}, R_p)$) cancel this OMC 1-specific contribution and isolate the size-dependent component, which is consistent across both OMC 1-included and OMC 1-excluded subsamples. OMC 1 is therefore best interpreted not as a counter-example to the size-dependent alignment picture, but as a high-radiation boundary condition on the physical parameter space within which that picture holds; a more comprehensive treatment of environment-stratified per-band polarization properties across the full POLSRC sample, including the LMSF, HMSF, and CMZ stratification, is given in companion Paper S2 (§3.4 and §3.5).

Requirements for Enhanced Models—Models that reproduce our observations would naturally incorporate: size-dependent alignment efficiencies where larger grains align more efficiently, as implemented in current RAT theory; environmental processing that modifies grain size distributions and optical properties; saturation behavior where alignment efficiency plateaus at high radiation intensities; and the ability to reproduce both the dual-wavelength-ratio observables ($\gamma - \alpha$, P_{450}/P_{850}) that are common to the sample, and the OMC 1-distinct per-band T_d - P and N_{H_2} - P behavior that emerges only in the highest-radiation regime probed here (Appendix B).

The robustness of $\gamma < \alpha$ across 75% of sources (21/28) provides a quantitative benchmark: dust models that cannot reproduce $\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ and the corresponding $P_{450}/P_{850} = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$ (per-source range 0.4–1.4) from first principles are missing important physics related to size-dependent alignment.

The Class 3 dust-emissivity and thermal degeneracies that survive this RAT-framed interpretation, together with the geometric (Class 1) and instrumental (Class 2) alternatives addressed elsewhere in this paper, are consolidated and discussed in §4.4.

4.4. Discriminating Power and Remaining Degeneracies

The observation $\gamma < \alpha$ in 75 % of the SPIDx sample, supported by $P_{450}/P_{850} < 1$ at the per-pixel level, constrains but does not uniquely determine the underlying physics. Our current data establish the phenomenology, while the most parsimonious interpretation—size-dependent grain alignment—requires confirmation from multiwavelength polarimetry extending to shorter (far-infrared extinction) or longer (millimetre emission) wavelengths. To clarify the boundary of what the present survey distinguishes, we organize candidate alternative mechanisms for $\gamma < \alpha$ into three classes: (Class 1) geometric and projection-induced explanations, (Class 2) instrumental and reduction-pipeline systematics, and (Class 3) dust-emissivity and thermal degeneracies. Classes 1 and 2 are addressed by the explicit analyses in §4.2 and in companion Paper T respectively, and are disfavored or bounded; Class 3 contains the genuine remaining degeneracy that requires future multi-wavelength data to resolve. Table 4 summarizes the Class 3 mechanisms together with discriminating observations that could break the degeneracy.

Class 1: Geometric Origins—Beam-averaging, line-of-sight magnetic-field tangling, polarization-weighted spatial mixing, and projection-induced depolarization could in principle bias the wavelength dependence of polarization observables. These geometric mechanisms are disfavored by the present results: Findings 1 and 3 jointly establish the wavelength-independent magnetic-field geometry and the geometry-independent P_{450}/P_{850} ratio (§4.1), and the explicit ruling-out analysis in §4.2 (Ruling Out Geometric Origins) shows that the surviving common-mode geometric depolarization within the OMC 1 high-radiation regime cancels in the wavelength-differential observables ($\gamma - \alpha$ and P_{450}/P_{850}). Geometric effects therefore cannot account for the survey-wide $\gamma < \alpha$ trend, although they may contribute to the source-to-source scatter through the residual common-mode component.

Class 2: Instrumental and Reduction Systematics—Spatial-frequency filtering mismatch between the 450 and 850 μm bands, absolute and relative flux calibration, beam-shape harmonization, and the additive 450 μm Stokes- Q and Stokes- U offset residual (companion Paper T §3.5, §3.1, and §3.3) are the principal reduction-level systematics relevant to dual-band polarization observables. Companion Paper T propagates the residual systematic budget into the inter-band uncertainties; the surviving budget is much smaller than the observed $\gamma - \alpha \simeq -0.5$ and the per-source P_{450}/P_{850} amplitude. Instrumental degeneracies are therefore bounded by the technical-validation analyses in companion Paper T rather than left as open systematic concerns.

Class 3: Dust-Emissivity and Thermal Degeneracies—The remaining degeneracies are physical rather than primarily instrumental. Line-of-sight temperature structure, optical-depth effects in the densest regions, and grain-composition variations can all modify the relation between total and polarized emission. However, Appendix C provides a simple two-temperature bounding calculation showing that line-of-sight temperature mixing alone gives $\gamma - \alpha = 0$ if the cold and warm components have the same polarization efficiency: in that limit PI_ν is simply proportional to I_ν and $\gamma = \alpha$ identically. Reproducing the observed negative $\gamma - \alpha$ therefore requires a component-dependent polarized-emissivity contrast, reinforcing the interpretation that the polarized and total emission are not tracing identical effective grain populations. The same cancellation holds for opacity-slope differences among mixed components if the polarized fraction is common. Producing the observed negative $\gamma - \alpha$ therefore requires a component-dependent polarized-emissivity or alignment-efficiency contrast. Within a two-temperature bounding calculation, the observed survey median $\gamma - \alpha \simeq -0.5$ is unreachable over plausible parameters unless the warm component is polarized far less efficiently than the cold component. Thus, line-of-sight structure may contribute to the source-to-source scatter, but it does not remove the need for a genuine difference between the grains or material dominating total emission and those dominating polarized emission.

Opacity-slope differences among line-of-sight components do not by themselves generate $\gamma - \alpha \neq 0$ if the polarized intensity remains a wavelength-independent fraction of the total intensity. Composition becomes relevant only insofar as the grains dominating polarized emission have a different effective opacity law from those dominating total emission.

Table 4. Class 3 Dust-Emissivity and Thermal Degeneracies in $\gamma < \alpha$

Mechanism	Predicted Signature	Discriminating Observation
Composition-dependent optical properties	Shallower polarized-emission spectrum	Far-IR extinction polarimetry
Line-of-sight temperature mixing	Warm and cold components contribute differently at 450 and 850 μm . A nonzero $\gamma - \alpha$ arises only if these components also have different effective polarization efficiencies or polarized-emissivity weights.	Appendix C shows that $\gamma - \alpha = 0$ identically when $p_w = p_c$ and that, over the plausible range $T_c = 10\text{--}20\text{ K}$, $T_w = 25\text{--}40\text{ K}$, $x_w \leq 0.5$, $p_w/p_c \geq 0.3$, the model does not reach the observed median $\gamma - \alpha \simeq -0.5$ (see Table 6).
Optical depth effects ($\tau \gtrsim 0.1$)	$\gamma - \alpha$ correlated with N_H	Optically thin tracers ($\lambda > 1\text{ mm}$)

NOTE—Three Class 3 mechanisms that can in principle produce $\gamma < \alpha$ within the dust-emissivity and thermal-structure regime, after Class 1 (geometric, §4.2) and Class 2 (instrumental, companion Paper T) explanations are excluded. The present survey constrains the phenomenology but does not uniquely distinguish between these scenarios; future multi-wavelength polarimetry will be required to break the remaining degeneracies.

Beyond temperature mixing, this compositional decoupling can itself generate $\gamma < \alpha$ through mixed-composition dust models. We briefly introduce the opacity-index notation used in this paragraph: β_P and β_I denote the opacity spectral indices of the polarized-intensity and total-intensity emission respectively, and β_{sil} and β_C denote those of pure silicate and carbonaceous grain populations. In the Rayleigh–Jeans limit valid for $h\nu \ll k_B T_d$ (e.g., Hildebrand 1983; Draine 2011), $\gamma - \alpha \simeq \beta_P - \beta_I$, so a simple picture in which only silicate grains are aligned while carbonaceous grains remain unaligned does not, by itself, produce $\gamma < \alpha$ for standard opacity indices ($\beta_{\text{sil}} \gtrsim \beta_C$). Instead, $\gamma < \alpha$ requires that the aligned grain population have a systematically flatter *effective* opacity spectral index than the bulk dust. This condition can be realized if the aligned grains are composite silicate–carbon aggregates whose optical properties differ from those of either constituent alone, for example through mixing, porosity, or coagulation/mantling during dust evolution (Köhler et al. 2015; Ysard et al. 2019; Guillet et al. 2018). In such models (e.g., Model D of Guillet et al. 2018, in which the aligned population consists of micron-sized astro-silicate grains coated with amorphous carbon mantles), the polarized emission selectively traces this aggregate population, whose submillimeter opacity spectrum is flatter than that of the carbon-rich small grains dominating the total-intensity SED, yielding a shallower polarized SED relative to the total emission. This compositional decoupling therefore provides an additional channel for $\gamma < \alpha$ beyond size-dependent alignment alone (Andersson et al. 2015).

Non-radiative alignment mechanisms, such as mechanical alignment of irregular or helical grains by gas flows (often referred to as “helical dust alignment”; Lazarian & Hoang 2007b; Hoang et al. 2018; Das & Weingartner 2016), could in principle maintain grain alignment in environments where the radiation field is weak. However, the OMC1-driven positive $P_{850}\text{--}T_d$ correlation in the full HGBS subsample favors RAT as the dominant mechanism in the highest-radiation regimes probed here (Appendix B). Additionally, cosmic-ray ionization, gas–grain drift, and iron inclusions (Draine & Lazarian 1999; Lazarian & Hoang 2008) may introduce secondary environmental dependencies on alignment efficiency. These processes are likely second-order modifiers rather than primary drivers of the systematic trends identified here. Disentangling the relative contributions requires multi-wavelength polarimetric observations spanning a wider wavelength range than is accessible with POL-2 alone.

Temperature mixing is not an alternative to wavelength-dependent polarized emissivity. It can only contribute to $\gamma - \alpha$ if the mixed components already have different polarization efficiencies or polarized-emissivity weights. Within plausible two-temperature parameters, even such a contrast cannot reproduce the observed median $\gamma - \alpha \simeq -0.5$ unless the contrast is extreme.

5. SUMMARY

We have presented a dual-wavelength submillimeter polarimetric analysis of the BISTRO 450/850 μm survey, focusing on the physical interpretation of size-dependent grain alignment. Our analysis is based on a statistical subsample of 28 sources with well-constrained spectral indices, consistent with the full POLSRC sample presented in the companion census paper (Paper S2) and built on the shared data-reduction and noise-modeling framework described in the

companion technical paper (Paper T). Three principal observational findings constrain submillimeter polarization physics:

1. *Wavelength-independent magnetic field orientation.* The polarization angles at 450 and 850 μm agree to within $|\chi_{450} - \chi_{850}|$ typically $\lesssim 10^\circ$ across the sample (12 of 13 targets have median $|\chi_{450} - \chi_{850}| \leq 10^\circ$; Monoceros R2, with median 11° , is the sole modest outlier), small compared to the 180° angular range of polarization. In the unified uncertainty notation introduced at §3, a typical pixel of the sample is described by $\Delta\chi = \bar{\Delta}\chi \pm 10^\circ_{\text{sys}} \pm 1.1^\circ_{\text{stat}}$ with $\bar{\Delta}\chi \lesssim 1^\circ$. The per-pixel statistical detection limit $\sigma_{\Delta\chi} \approx 1.1^\circ$ resolves small but significant offsets in a few sources—with Monoceros R2 the most prominent—though these are physically minor and do not affect the principal findings. Both wavelengths predominantly trace the same plane-of-sky magnetic field structure, indicating that any wavelength dependence in polarization fraction is unlikely to be primarily driven by geometric effects such as line-of-sight field variations.
2. *Systematic $\gamma < \alpha$ across diverse environments.* The polarized-intensity spectral index γ is smaller than the total-intensity spectral index α in 21 of 28 sources (75%). Per-source medians are $\langle\alpha\rangle = 3.46 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ and $\langle\gamma\rangle = 3.12 \pm 0.04_{\text{sys}} \pm 0.11_{\text{stat}}$, where the statistical errors are empirical SEMs from the per-source dispersion of the SPIDX sample. These yield the per-pixel median difference $\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ and a characteristic polarization ratio $P_{450}/P_{850} = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$, with per-source values spanning 0.4–1.4. This systematic difference is observed across low-mass cores (e.g., Ophiuchus A), intermediate-mass regions (e.g., OMC 1), high-mass star-forming regions (e.g., NGC 6334), and Galactic Center clouds (e.g., Sagittarius B2). *The implication is that polarized emission is dominated by well-aligned grains whose effective spectral dependence is shallower than that of the bulk dust emission.*
3. *Geometry-independent $P_{450}/P_{850} < 1$.* The polarization-fraction ratio shows no systematic dependence on the inter-band angle difference $|\chi_{450} - \chi_{850}|$, and a modest but statistically significant negative ranked correlation between polarization fraction and the polarization-angle dispersion function $|\mathcal{S}|$ is observed at both wavelengths within the above-floor sample ($r_s = -0.19$, $N = 345$, $p = 5.0 \times 10^{-4}$ at 450 μm ; $r_s = -0.14$, $N = 372$, $p = 5.7 \times 10^{-3}$ at 850 μm ; §3.2), driven primarily by the OMC 1 high-radiation regime and collapsing to statistical insignificance outside it. Used here as a presence test for geometric depolarization within the OMC 1 regime rather than as a survey-wide quantitative measure, this confirms that geometric depolarization operates at both 450 and 850 μm where it is measurable, yet its common contribution cancels in the ratio. Combined with Finding 1, this isolates intrinsic grain physics—most naturally size-dependent alignment efficiency—as the origin of the wavelength-dependent polarization fraction, rather than geometric artefacts.

These three findings directly address the open questions posed in Section 1: Finding 1 establishes that 450 and 850 μm polarization trace a common plane-of-sky magnetic field geometry, so wavelength differences in polarization fraction are not driven by geometric effects; Finding 2 demonstrates that the polarized-intensity spectral index γ systematically falls below the total-intensity spectral index α across diverse environments; and Finding 3 shows that the $P_{450}/P_{850} < 1$ signature is geometry-independent, isolating intrinsic grain physics as its origin.

Together, these results provide a coherent body of evidence most consistent with size-dependent grain alignment in star-forming regions. Our findings provide quantitative benchmarks— $\gamma - \alpha = -0.52 \pm 0.04_{\text{sys}} \pm 0.10_{\text{stat}}$ and the sample-median $P_{450}/P_{850} = 0.67 \pm 0.02_{\text{sys}} \pm 0.04_{\text{stat}}$ (per-source range 0.4–1.4)—that theoretical dust models and magnetohydrodynamic simulations must reproduce within their combined systematic and statistical budgets. The *universality-level* extension of these existence-level findings is presented in companion [Paper S2](#) as findings F4–F6: F4 establishes the population-level robustness of F1–F3 across the BISTRO POLSRC sample, F5 demonstrates regime-dependent wavelength response, with the median $\gamma - \alpha$ and R_p both increasing from LMSF through HMSF to CMZ, and F6 quantifies disorder-mediated wavelength divergence under local N_H and T_d coupling within the Herschel Gould Belt Survey 9-field subsample. The unified F1–F6 set thus partitions naturally into the existence layer (F1–F3, this paper) and the universality layer (F4–F6, Paper S2), and constitutes the joint observational benchmark that any candidate grain-alignment mechanism must reproduce.

A two-temperature line-of-sight mixing test (§4.4) shows that temperature mixing alone cannot reproduce the observed median $\gamma - \alpha \simeq -0.5$: if the cold and warm components have the same polarization efficiency, $\gamma = \alpha$ identically, and plausible contrasts in p_w/p_c do not reach the observed value. The negative $\gamma - \alpha$ therefore requires a genuine difference between the effective grain populations or polarized-emissivity weights traced by total and polarized emission.

Beyond the benchmark numerics themselves, the spectral-index difference $\gamma - \alpha$ (equivalently, P_{450}/P_{850}) offers a distance-independent diagnostic of grain physics; the subset of sources with $\gamma > \alpha$ may represent distinct physical conditions and warrant targeted follow-up, though this diagnostic has not yet been calibrated against independent evolutionary indicators and is therefore to be regarded as a hypothesis rather than an established metric. We also note that the sample is heterogeneous in distance (139 pc to 8.3 kpc), spatial resolution, and evolutionary state — caveats that qualify the interpretation but do not weaken the model-independent nature of the three findings.

Methodologically, this work demonstrates that by employing the polarization ratio P_{450}/P_{850} , the spectral-index difference $\gamma - \alpha$, and the geometry-controlled P – S relation, one can extract size-dependent grain-alignment information indirectly—without requiring explicit normalisation by dust column density or extinction. Beyond the specific BISTRO SPIDX subsample analysed here, the present analysis highlights that the three wavelength-differential quantities R_p , $\gamma - \alpha$, and $|\chi_{450} - \chi_{850}|$ form a mutually complementary diagnostic set: the first traces the net polarization-spectrum amplitude, the second traces the differential emissivity behavior between polarized and total dust emission, and the third controls for geometric magnetic-field mismatch. Together they provide an observationally economical but physically powerful toolkit for interpreting future submillimeter polarimetric surveys, playing a role for emission polarimetry analogous to that of p_V/A_V in optical and infrared extinction studies.

Complementary per-source deliverables—polarization maps, the polarization-angle dispersion function $\mathcal{S}$, the alignment-efficiency proxy $\mathcal{S} \times \mathcal{P}^{\text{frac}}$, spectral-index catalogues, and dual-wavelength magnetic-field orientation atlases—are published separately for all 69 sources in the companion census paper S2, which additionally stratifies environment-dependent analyses across low- and intermediate-mass star-forming regions, high-mass star-forming regions, and the Central Molecular Zone. Technical documentation of the noise model, 450 μm offset correction, and independent SOFIA/HAWC+ validation is available in Paper T.

Claim-Strength Hierarchy—The principal results of this paper sort into four claim-strength tiers, ordered by the degree to which they are anchored to direct observation. *Directly measured* from the matched-resolution dual-band BISTRO data are the per-pixel polarization-angle differences $|\chi_{450} - \chi_{850}|$, the per-source spectral indices α and γ on the SPIDX subsample, and the per-pixel and per-source polarization-fraction ratios R_p . *Empirically inferred* from those measurements are the F1–F3 universality statements: inter-band field-angle agreement to within $|\Delta\chi| \lesssim 4^\circ\text{--}6^\circ$ across the 21-source dual-band-angle sample, $\gamma < \alpha$ in 21 of 28 sources at the systematic-floor-controlled level, and $R_p < 1$ in roughly four-fifths of the survey sources. *Physically interpreted* as a size-dependent grain-alignment signature is the joint $\gamma < \alpha$, $R_p < 1$, and geometric-independence pattern under a RAT-consistent two-population dust model in which the polarization-weighted grains exhibit a shallower effective emissivity spectrum than the bulk. *Not yet uniquely determined* from these observations alone, and explicitly acknowledged as residual degeneracies in §4.4, are: the precise grain-population identification beyond the size-dependent framing, the relative roles of opacity-slope mixing and component-dependent polarized emissivity for the observed $\gamma - \alpha < 0$, and the per-source magnetic-disorder geometry of individual stress-test cases such as Mon R2.

All per-source diagnostic products described in this paper, including the dual-band polarization-angle catalogues, the spectral-index per-source statistics, and the polarization-fraction-ratio measurements summarised in Table 2, will be made available electronically as supplementary material to this paper.

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APPENDIX

A. ASSESSMENT OF $^{12}\text{CO}(J=3-2)$ CONTAMINATION IN THE POL-2 850 μm BAND

The SCUBA-2/POL-2 850 μm continuum band encompasses the $^{12}\text{CO}(J=3-2)$ transition at 345.796 GHz. Strong $^{12}\text{CO}(J=3-2)$ line emission can therefore, in principle, contribute to the measured continuum intensity. Because this effect is expected to be most pronounced in regions of weak dust emission, such as molecular outflows or cloud edges, we explicitly evaluated the level of $^{12}\text{CO}(J=3-2)$ contamination in our Orion A dataset.

Data and Preprocessing—We used the $^{12}\text{CO}(J=3-2)$ integrated intensity map obtained with JCMT/HARP, expressed in units of K km s^{-1} (Buckle et al. 2012). Following standard practice, the map was first corrected for the HARP main-beam efficiency ($\eta_{\text{MB}} = 0.61$) to convert antenna temperatures to main-beam temperatures. The corrected $^{12}\text{CO}(J=3-2)$ map was then reprojected onto the POL-2 850 μm pixel grid using bilinear interpolation.

To ensure a fair comparison with the ground-based POL-2 continuum data, we uniformly applied a SCUBA-2-like spatial high-pass filter to the $^{12}\text{CO}(J=3-2)$ map across the entire mosaicked region. Specifically, spatial structures larger than $\sim 300''$ were removed by subtracting a Gaussian-smoothed version of the map. This step mimics the spatial-frequency filtering inherent to SCUBA-2 data reduction and prevents large-scale $^{12}\text{CO}(J=3-2)$ emission, which is not recovered in POL-2 maps, from artificially inflating the inferred contamination level.

Conversion from $^{12}\text{CO}(J=3-2)$ Integrated Intensity to Continuum Flux—To estimate the contribution of $^{12}\text{CO}(J=3-2)$ line emission to the 850 μm continuum band, we followed the methodology developed by Drabek et al. (2012). In

this approach, the integrated $^{12}\text{CO}(J=3-2)$ intensity is converted to an equivalent continuum flux density using a bandpass-averaged conversion factor that accounts for the dilution of a narrow spectral line within the broad SCUBA-2 continuum band.

We adopted a conversion coefficient of $C_{850} = 0.63 \text{ mJy beam}^{-1} \text{ per K km s}^{-1}$, corresponding to Grade 1 weather conditions ($\tau_{225} < 0.05$) as tabulated in Drabek et al. (2012, Table 1). This value represents the most conservative estimate (i.e., the lowest conversion factor), providing a lower bound on the expected $^{12}\text{CO}(J=3-2)$ contamination. Under typical observing conditions (Grades 2–3), the conversion factor would be 8–11% higher ($0.68\text{--}0.70 \text{ mJy beam}^{-1} \text{ per K km s}^{-1}$).

We note that this conversion factor was derived for the SCUBA-2 continuum bandpass without the POL-2 module. The effective POL-2 bandpass, including the half-wave plate and wire-grid analyzer, has not been independently characterized for molecular line contamination. We therefore adopt this conversion as a best-effort estimate and explicitly treat the resulting contamination levels as approximate rather than exact.

The $^{12}\text{CO}(J=3-2)$ contribution was first computed in units of mJy beam^{-1} and then converted to mJy arcsec^{-2} using the POL-2 850 μm beam area, $\Omega_{\text{beam}} = 1.133 \text{ FWHM}^2$ with $\text{FWHM} = 12''.6$ (Mairs et al. 2021). The POL-2 Stokes I map was verified to be in units of mJy arcsec^{-2} , ensuring consistent comparison. We note that the POL-2 Stokes I intensity near map edges is typically lower than the corresponding SCUBA-2 total intensity due to reduced integration time and coverage in the POL-2 scan pattern; consequently, the contamination ratio R_{CO} computed here may overestimate the true contamination level in these peripheral regions.

Contamination Ratio and Pixel Selection—The $^{12}\text{CO}(J=3-2)$ contamination ratio, $R_{\text{CO}} \equiv I_{\text{CO}}/I_{850}$, was defined as the ratio of the estimated $^{12}\text{CO}(J=3-2)$ -derived continuum contribution to the observed POL-2 Stokes I intensity at each pixel. Pixels with non-finite values or negative intensities were excluded from the analysis.

For the statistical assessment most relevant to the polarization analysis, we further restricted the sample to science-quality pixels satisfying a signal-to-noise ratio threshold of $I_{850}/\Delta I_{850} > 15$, where ΔI_{850} was derived from the variance extension of the POL-2 FITS files. This threshold is consistent with the intensity cuts adopted for polarization segment selection throughout the main text.

Figure 7 presents the resulting $^{12}\text{CO}(J=3-2)$ contamination ratio map at 850 μm for the mosaicked POL-2 fields in Orion A, covering the region from OMC 1 through OMC 4. The spatial coverage corresponds to the individual field maps presented in companion Paper S2 (Paper S2 Orion A atlas). Pixels failing the signal-to-noise criterion were masked. The map is displayed on a logarithmic scale to emphasize the large dynamic range of the contamination ratio.

Statistical Characterization—Because the contamination ratio spans several orders of magnitude, and to assess the robustness of the contamination estimate across the full mosaicked region, its distribution was characterized using $\log_{10}(R_{\text{CO}})$. For science-quality pixels satisfying $I_{850}/\Delta I_{850} > 15$, the median contamination level is $\log_{10}(R_{\text{CO}}) = -1.01$, corresponding to a contamination fraction of $\sim 10\%$. The central 90% of the distribution (5th to 95th percentile) spans $\log_{10}(R_{\text{CO}}) \simeq -2.00$ to -0.30 , equivalent to contamination fractions of approximately 1% to 50%. Approximately 5% of the pixels have contamination below 1%, 30% fall below 5%, and 51% lie below 10%.

When weighted by Stokes I intensity, the effective contamination level is lower, with an intensity-weighted mean of $\sim 8\%$. The intensity-weighted mean provides a more representative measure of contamination for polarimetric analyses, because brighter pixels dominate the polarized flux and thus have greater influence on the derived magnetic field statistics. This lower weighted value reflects the fact that the highest contamination fractions occur preferentially in low-intensity regions, which contribute relatively little to the polarized emission budget and to the derived magnetic field statistics.

Implications for the Polarization Analysis—Even in regions where the $^{12}\text{CO}(J=3-2)$ contribution approaches the 10% level, the inferred contamination remains comparable to or below the typical absolute calibration uncertainty of SCUBA-2/POL-2 continuum observations, which is of order 10–15% (Mairs et al. 2021). Moreover, the polarization analysis presented in this paper focuses on dense, high signal-to-noise regions where the relative $^{12}\text{CO}(J=3-2)$ contribution is systematically smaller and the dust continuum emission dominates.

We note that the SCUBA-2/POL-2 850 μm band includes the $^{12}\text{CO}(J=3-2)$ line, and that polarized line emission via the Goldreich–Kylafis (GK) effect may in principle contribute to the observed polarization signal under specific physical conditions. In particular, the GK effect is expected to be efficient only when the line optical depth is of order unity and the velocity field is sufficiently coherent, such that anisotropic radiative excitation dominates over collisional depolarization (e.g., Goldreich & Kylafis 1981, 1982; Cortes et al. 2005; Ching et al. 2016). In the present

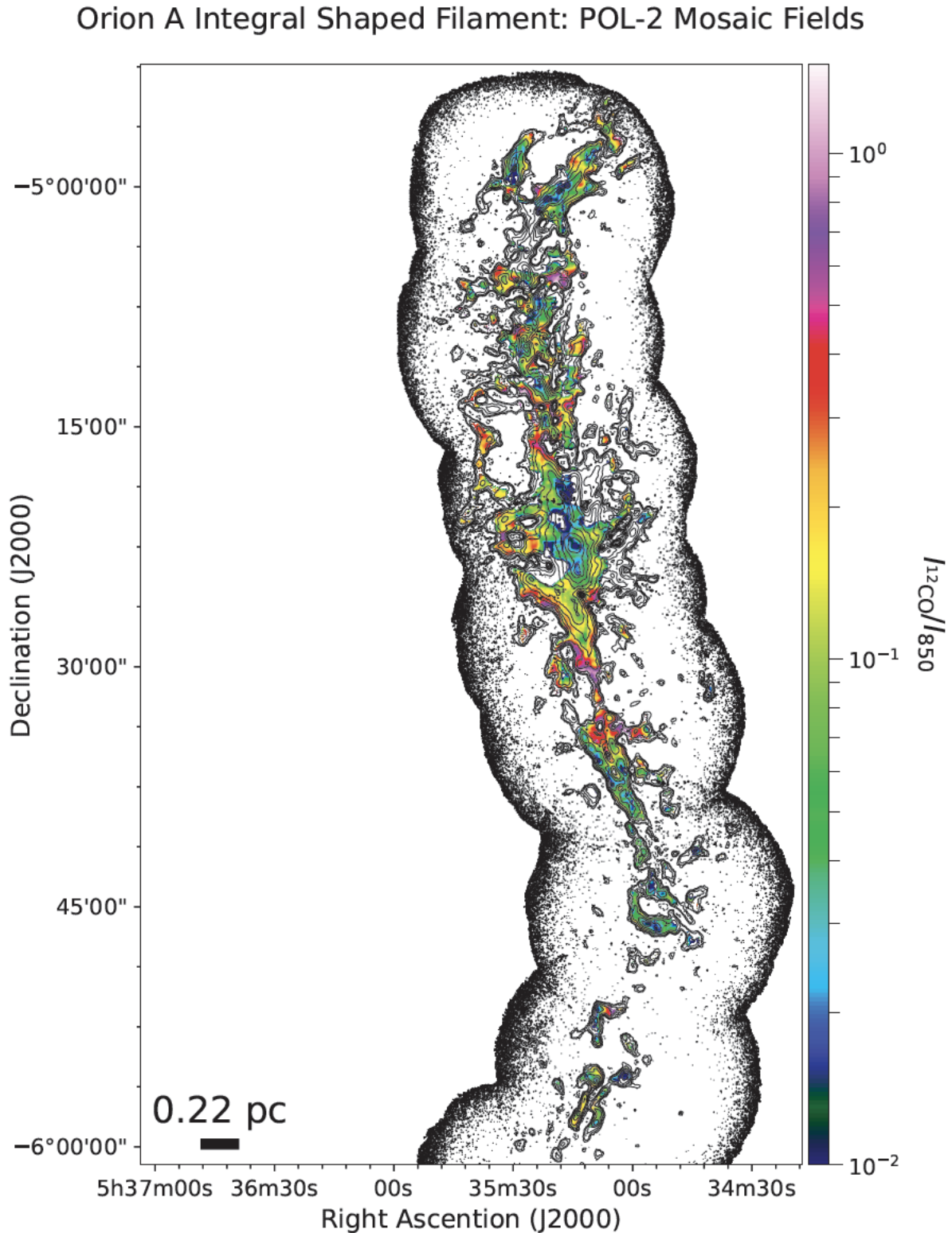


Figure 7. Map of the $^{12}\text{CO}(J=3-2)$ contamination ratio $R_{\text{CO}} \equiv I_{\text{CO}}/I_{850}$ at $850\ \mu\text{m}$ for the mosaicked POL-2 fields in Orion A. The spatial coverage extends from OMC 1 through OMC 4. Only pixels satisfying $I_{850}/\Delta I_{850} > 15$ are shown. The color scale is logarithmic, and the overlaid contours trace the POL-2 Stokes I emission. The highest contamination fractions occur preferentially in low-intensity regions surrounding dense filaments.

study, any such contamination is expected to be negligible for the global statistical results, as the polarized emission is overwhelmingly dominated by dust continuum emission and our conclusions are based on ensemble properties rather than localized features.

Our findings are consistent with the comprehensive study of $^{12}\text{CO}(J=3-2)$ contamination in Orion A by Coudé et al. (2016), who found that contamination levels exceeding 10% are confined to regions with 850 μm flux densities below $\sim 300 \text{ mJy beam}^{-1}$. Their analysis demonstrated that the optically thick $^{12}\text{CO}(J=3-2)$ line saturates more rapidly than the dust continuum, causing contamination to become significant primarily in low column density regions such as molecular outflows and cloud edges—precisely the regions excluded by our intensity-based pixel selection criteria.

We therefore conclude that contamination from $^{12}\text{CO}(J=3-2)$ emission does not significantly affect the polarization fractions or derived quantities discussed in the main text, and no explicit correction for $^{12}\text{CO}(J=3-2)$ line contamination has been applied.

Note on $^{12}\text{CO}(J=6-5)$ Contamination at 450 μm .—We did not attempt to evaluate potential contamination of the SCUBA-2 450 μm continuum band by the $^{12}\text{CO}(J=6-5)$ transition at 691.473 GHz. No suitable wide-field $^{12}\text{CO}(J=6-5)$ data are available for Orion A, and inferring its contribution from lower- J transitions would require assumptions about excitation conditions that introduce large uncertainties. Additionally, the $^{12}\text{CO}(J=6-5)$ line lies near the edge of the SCUBA-2 450 μm bandpass, where the instrumental transmission is low compared to the band center (Drabek et al. 2012). Under good atmospheric conditions (JCMT weather band 1), the atmospheric transmission at these frequencies improves, but the $^{12}\text{CO}(J=6-5)$ conversion factor remains comparable to that of $^{12}\text{CO}(J=3-2)$ at 850 μm ($\sim 0.64 \text{ mJy beam}^{-1}$ per K km s^{-1} for Grade 1; Drabek et al. 2012). Given that the 450 μm dust continuum flux is typically a factor of 6–12 higher than at 850 μm for the dust temperatures in our targets, any $^{12}\text{CO}(J=6-5)$ contamination is expected to be proportionally smaller than the $^{12}\text{CO}(J=3-2)$ contamination at 850 μm . For these reasons, $^{12}\text{CO}(J=6-5)$ line contamination at 450 μm is not considered further in this study. In summary, the 850 μm band is more susceptible to CO line contamination than the 450 μm band: at 850 μm , the $^{12}\text{CO}(J=3-2)$ line falls near the band center with a conversion factor of $\sim 0.63 \text{ mJy beam}^{-1}$ per K km s^{-1} (Drabek et al. 2012), whereas at 450 μm , the $^{12}\text{CO}(J=6-5)$ line lies at the band edge where instrumental transmission is low, and the higher dust continuum flux further dilutes any line contribution. Moreover, the 450 μm data in this study were taken exclusively under Weather Band 1 conditions ($\tau_{225}^{\text{zenith}} \leq 0.05$), under which atmospheric transmission at the CO($J=6-5$) frequency improves but the line-to-continuum ratio remains negligible.

B. PER-PIXEL ENVIRONMENTAL CORRELATIONS: OMC 1-EXCLUDED VS FULL HGBS SUBSAMPLE

The discussion in §4.3 refers to a set of per-pixel Spearman rank correlations between polarization fraction at each wavelength and the local dust temperature T_d and column density N_{H_2} (HGBS), whose magnitude and sign depend on whether OMC 1—the warmest and most luminous single source in the HGBS subsample—is included. Table 5 reports the full numerical comparison, together with the $I_{450}/I_{850}-P_{450}/P_{850}$ anticorrelation used in the main text (§3.3, Figure 5) as a Tier-1-grade dual-wavelength observable that is preserved under OMC 1 exclusion.

C. A TWO-TEMPERATURE LINE-OF-SIGHT MIXING TEST

Following the discussion of §4.2, we examine whether line-of-sight temperature mixing can reproduce the observed survey median $\gamma - \alpha \simeq -0.5$ (§3.3; Paper S2 Table 7), and quantify how large a component-dependent polarization-efficiency contrast would be required if it could. We adopt a two-component line-of-sight emission model in which a cold component at temperature T_c contributes a mass fraction $1 - x_w$ and a warm component at T_w contributes x_w , both with a common modified-blackbody opacity index β . The total intensity at frequency ν is

$$I_\nu \propto \nu^\beta [(1 - x_w) B_\nu(T_c) + x_w B_\nu(T_w)], \quad (\text{C1})$$

and the polarized intensity, with p_c and p_w representing the effective polarization efficiencies of the cold and warm components, is

$$PI_\nu \propto \nu^\beta [p_c (1 - x_w) B_\nu(T_c) + p_w x_w B_\nu(T_w)]. \quad (\text{C2})$$

The dual-band spectral indices ($\nu_{450} = 666 \text{ GHz}$, $\nu_{850} = 353 \text{ GHz}$; central SCUBA-2/POL-2 frequencies as in companion Paper T §3.1 and the body of this paper) are computed as $\alpha = \log(I_{450}/I_{850})/\log(\nu_{450}/\nu_{850})$ and

Table 5. Per-Pixel Environmental Correlations: OMC 1-Excluded vs Full HGBS Subsample

Pair	r_s (excl OMC 1)	N	p (excl OMC 1)	r_s (full)	N	p (full)
T_d, P_{450}	-0.32	286	2.5×10^{-8}	+0.09	348	8.5×10^{-2}
T_d, P_{850}	+0.03	321	5.5×10^{-1}	+0.36	383	4.4×10^{-13}
N_{H_2}, P_{450}	-0.49	286	2.6×10^{-18}	-0.11	348	4.5×10^{-2}
N_{H_2}, P_{850}	-0.13	321	1.7×10^{-2}	+0.21	383	3.5×10^{-5}
$I_{450}/I_{850}, P_{450}/P_{850}$	-0.35	267	4.4×10^{-9}	-0.49	329	1.9×10^{-21}

NOTE—Spearman rank correlations for the 9-source HGBS subsample (Method 3, per-pixel detection mask $P_i \geq 5 \delta P_i$). “Full” refers to all qualifying pixels in the HGBS subsample; “excl OMC 1” refers to the same subsample with the 62 pixels of OMC 1 removed. The first four pairs (per-band T_d – P and N_{H_2} – P) show sign-flips or large magnitude changes between the two samples and are sensitive to the OMC 1 high-radiation boundary of the physical parameter space sampled here (§4.3). The dual-wavelength ratio pair $I_{450}/I_{850}, P_{450}/P_{850}$ is preserved in both samples and is treated in §3.3 as a per-pixel manifestation of the $\gamma - \alpha$ spectral-index structure. Per-pixel statistical detection limits at the adopted $\sigma = 5$ threshold are propagated through the floor-mask construction in companion Paper T §3.3.

$\gamma = \log(P_{I_{450}}/P_{I_{850}})/\log(\nu_{450}/\nu_{850})$. Because the ν^β opacity factor is common to numerator and denominator at each frequency, β cancels analytically from $\gamma - \alpha$ within the same- β assumption: the difference depends only on T_c , T_w , x_w , and the polarization-efficiency contrast $r \equiv p_w/p_c$. Here r is the ratio of the *effective* polarization efficiencies assigned to the warm and cold line-of-sight components within the toy model, and is not the observed dual-band polarization-fraction ratio $R_p \equiv P_{450}/P_{850}$ (§3.2).

Adopted temperature range—The temperature ranges used in this bounding calculation are chosen to bracket the representative dust temperatures adopted for the spectral-index analysis in companion Paper S2 §3.4.1. In that analysis, low-mass star-forming regions and cold infrared dark clouds are represented by $T_d = 15 \pm 5$ K, while high-mass star-forming and CMZ sources are represented by $T_d = 25 \pm 10$ K, with HGBS-derived temperatures used where available. We therefore take $T_c = 10$ –20 K as the cold component range and $T_w = 25$ –40 K as a deliberately broad warm component range. These intervals are not intended as source-by-source temperature measurements; rather, they define a conservative parameter space for testing whether plausible line-of-sight temperature mixtures can mimic the observed median $\gamma - \alpha \simeq -0.5$. The upper end of the warm range extends beyond the nominal 25 ± 10 K default to make the test conservative: warmer components increase the 450/850 contrast and therefore maximize the ability of temperature mixing to drive $\gamma - \alpha$ away from zero.

Limiting case: $p_w = p_c$ —A useful limiting case is $p_w = p_c$ ($r = 1$). In this case $PI_\nu = p_c I_\nu$ at both wavelengths, and therefore $\gamma = \alpha$ identically, independent of T_c , T_w , x_w , and the adopted opacity index β . We verified this numerically to machine precision over $\beta = 1.5$ –3.0. The same cancellation holds even when the cold and warm components have different opacity slopes ($\beta_c \neq \beta_w$), provided the polarization efficiency is common to both components. Thus, neither temperature mixing nor opacity-slope mixing alone produces a nonzero $\gamma - \alpha$; a component-dependent polarized-emissivity or polarization-efficiency contrast ($r \neq 1$) is required.

Parameter-scan results—Figure 8(a) shows $\gamma - \alpha$ as a function of r and x_w at the representative temperature pair $T_c = 15$ K, $T_w = 30$ K, with $\beta = 2.23$ (the median β of the spectral-index sample, Paper T §3.1); the vertical dashed line at $r = 1$ marks the cancellation locus and the dotted line at $r = 0.3$ marks the plausible lower bound adopted below. Figure 8(b) shows a representative conservative slice over the cold and warm dust temperatures at fixed $x_w = 0.30$ and $r = 0.3$. These panels are illustrative slices, not the full parameter scan.

The full grid search over the plausible range ($T_c = 10$ –20 K, $T_w = 25$ –40 K, $x_w \leq 0.5$, and $r = p_w/p_c \geq 0.3$, 32 912 grid points) confirms that the observed survey median $\gamma - \alpha \simeq -0.5$ is *not* reached anywhere: the most negative value in this range is $\gamma - \alpha = -0.354$ (at $T_c = 10$ K, $T_w = 40$ K, $x_w = 0.15$, $r = 0.31$), and the fraction of plausible models reaching $\gamma - \alpha \leq -0.5$ is zero. Restricting further to the moderate-contrast subset $0.5 \leq r \leq 2$ (19 360 grid points) the fraction also remains zero. The target value -0.5 can be reached only in the wider mathematical grid that allows

r as small as 0.05, where the warm component is polarized one to two orders of magnitude less efficiently than the cold; in the wider grid the most negative attainable value is $\gamma - \alpha = -0.81$ (at $T_c = 10\text{ K}$, $T_w = 40\text{ K}$, $x_w = 0.25$, $r = 0.05$), and 1,929 of 58,080 points (3.3%) reach $\gamma - \alpha \leq -0.5$. Models that do reach the observed magnitude (Table 6) therefore require $r \lesssim 0.15$, an extreme polarization-efficiency contrast outside our adopted plausible range. Line-of-sight temperature mixing is therefore not a viable explanation for the observed negative $\gamma - \alpha$ within physically reasonable parameters; it can contribute only if accompanied by a strong component-dependent polarized-emissivity or alignment-efficiency contrast.

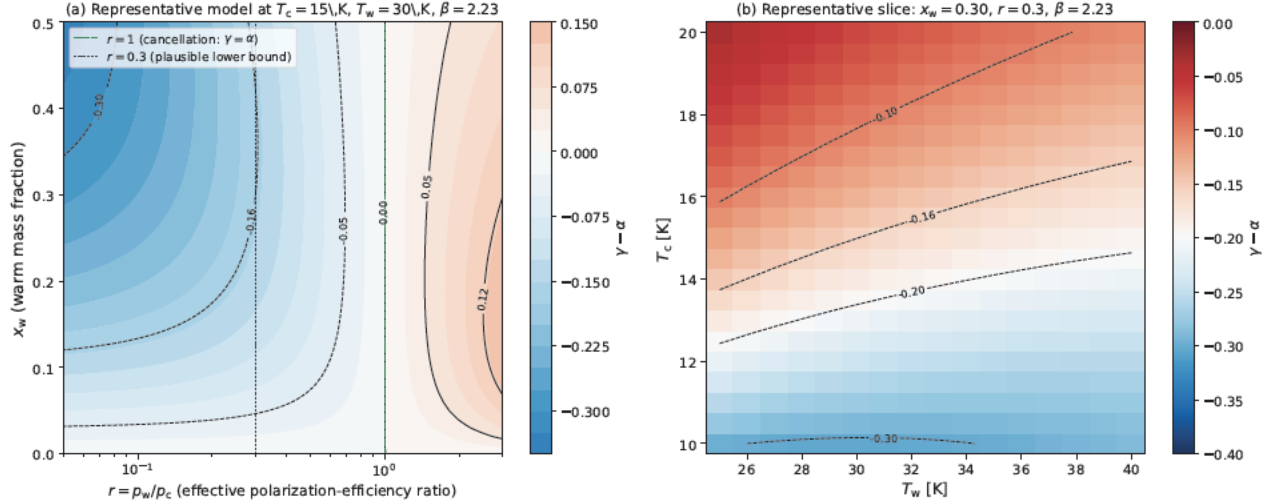


Figure 8. Two-temperature line-of-sight mixing test for the spectral-index difference $\gamma - \alpha$. (a) Representative model at $T_c = 15\text{ K}$, $T_w = 30\text{ K}$, and $\beta = 2.23$, shown as a function of the warm mass fraction x_w and the polarization-efficiency ratio $r = p_w/p_c$. The green dashed vertical line at $r = 1$ marks the cancellation limit in which PI_ν is proportional to I_ν and therefore $\gamma = \alpha$ identically; the black dotted line at $r = 0.3$ marks the plausible lower bound adopted in the full grid scan. Negative $\gamma - \alpha$ requires $r < 1$, i.e., the warm component must be less efficiently polarized than the cold component. The observed median $\gamma - \alpha \simeq -0.5$ lies outside the plotted range for this representative temperature pair (the most negative value reached at $T_c = 15\text{ K}$, $T_w = 30\text{ K}$ is $\gamma - \alpha \approx -0.34$ at the lower edge $r = 0.05$, $x_w = 0.5$). (b) Representative conservative slice at $x_w = 0.30$ and $r = 0.3$, shown over the cold and warm dust temperatures. Even in this low- r case, the observed survey median $\gamma - \alpha \simeq -0.5$ is not reached anywhere in the plotted (T_c, T_w) plane. The full grid search confirms that no model in the adopted plausible range ($T_c = 10\text{--}20\text{ K}$, $T_w = 25\text{--}40\text{ K}$, $x_w \leq 0.5$, and $r \geq 0.3$) reaches the observed median (see body text and Table 6). Here $r = p_w/p_c$ is the ratio of the effective polarization efficiencies assigned to the warm and cold line-of-sight components within the toy model, and is *not* the observed dual-band polarization-fraction ratio $R_p \equiv P_{450}/P_{850}$ of §3.2.

Table 6. Polarization-efficiency contrast required to reproduce the observed median $\gamma - \alpha \simeq -0.5$ in the two-temperature toy model. “Unreachable” indicates that the target value is not reached even in the limit $p_w/p_c \rightarrow 0$ for the specified temperatures and warm mass fraction. Numerical inversion uses `scipy.optimize.brentq` on $r \in [10^{-4}, 1]$.

T_c [K]	T_w [K]	x_w	Required p_w/p_c
15	30	0.30	unreachable
15	30	0.50	unreachable
10	30	0.30	0.149
10	40	0.50	0.092
12	25	0.40	0.022

Connection to the Class 3 degeneracies of §4.4—This calculation refines the Class 3 “Temperature mixing” degeneracy of Table 4: opacity-slope or temperature differences *among* line-of-sight components do not by themselves generate $\gamma - \alpha \neq 0$ if the polarized fraction is the same for all components. Composition becomes relevant in $\gamma - \alpha$ only insofar as the grains dominating polarized emission have a *different effective opacity law* from those dominating total

emission—a scenario *not* captured by the same-component, common-polarization-efficiency toy model adopted here, and which therefore remains a real Class 3 degeneracy in §4.4. The toy-model conclusion is that within-component temperature mixing alone is insufficient to reproduce the observed $\gamma - \alpha$; resolving the residual ambiguity between component-dependent emissivity contrasts and component-dependent grain populations requires multi-band polarimetric or laboratory dust-emissivity constraints beyond the scope of the present analysis.

REFERENCES

- Alves, F. O., Frau, P., Girart, J. M., et al. 2014, *A&A*, 569, L1, doi: [10.1051/0004-6361/201424678](https://doi.org/10.1051/0004-6361/201424678)
- Andersson, B. G., Lazarian, A., & Vaillancourt, J. E. 2015, *Annual Review of Astronomy and Astrophysics*, 53, 501, doi: [10.1146/annurev-astro-082214-122414](https://doi.org/10.1146/annurev-astro-082214-122414)
- André, P., Men'shchikov, A., Bontemps, S., et al. 2010, *A&A*, 518, L102, doi: [10.1051/0004-6361/201014666](https://doi.org/10.1051/0004-6361/201014666)
- Buckle, J. V., Davis, C. J., di Francesco, J., et al. 2012, *MNRAS*, 422, 521, doi: [10.1111/j.1365-2966.2012.20628.x](https://doi.org/10.1111/j.1365-2966.2012.20628.x)
- Chapin, E. L., Berry, D. S., Gibb, A. G., et al. 2013, *Monthly Notices of the Royal Astronomical Society*, 430, 2545, doi: [10.1093/mnras/stt052](https://doi.org/10.1093/mnras/stt052)
- Ching, T.-C., Lai, S.-P., Zhang, Q., et al. 2016, *ApJ*, 819, 159, doi: [10.3847/0004-637X/819/2/159](https://doi.org/10.3847/0004-637X/819/2/159)
- Cortes, P. C., Crutcher, R. M., & Watson, W. D. 2005, *ApJ*, 628, 780, doi: [10.1086/430815](https://doi.org/10.1086/430815)
- Coudé, S., Bastien, P., Kirk, H., et al. 2016, *MNRAS*, 457, 2139, doi: [10.1093/mnras/stv3009](https://doi.org/10.1093/mnras/stv3009)
- Coudé, S., Bastien, P., Houde, M., et al. 2019, *The Astrophysical Journal*, 877, 88, doi: [10.3847/1538-4357/ab1b23](https://doi.org/10.3847/1538-4357/ab1b23)
- Crutcher, R. M. 2012, *ARA&A*, 50, 29, doi: [10.1146/annurev-astro-081811-125514](https://doi.org/10.1146/annurev-astro-081811-125514)
- Das, I., & Weingartner, J. C. 2016, *MNRAS*, 457, 1958, doi: [10.1093/mnras/stw146](https://doi.org/10.1093/mnras/stw146)
- Davis, Jr., L., & Greenstein, J. L. 1951, *The Astrophysical Journal*, 114, 206, doi: [10.1086/145464](https://doi.org/10.1086/145464)
- Drabek, E., Hatchell, J., Friberg, P., et al. 2012, *MNRAS*, 426, 23, doi: [10.1111/j.1365-2966.2012.21140.x](https://doi.org/10.1111/j.1365-2966.2012.21140.x)
- Draine, B. T. 2003, *ARA&A*, 41, 241, doi: [10.1146/annurev.astro.41.011802.094840](https://doi.org/10.1146/annurev.astro.41.011802.094840)
- . 2011, *Physics of the Interstellar and Intergalactic Medium*, Princeton Series in Astrophysics (Princeton, NJ: Princeton University Press)
- Draine, B. T., & Lazarian, A. 1999, *ApJ*, 512, 740, doi: [10.1086/306809](https://doi.org/10.1086/306809)
- Fanciullo, L., Guillet, V., Boulanger, F., & Jones, A. P. 2022, *ApJ*, 928, 92, doi: [10.3847/1538-4357/ac5039](https://doi.org/10.3847/1538-4357/ac5039)
- Fissel, L. M., Ade, P. A. R., Angilè, F. E., et al. 2016, *ApJ*, 824, 134, doi: [10.3847/0004-637X/824/2/134](https://doi.org/10.3847/0004-637X/824/2/134)
- Friberg, P., Bastien, P., Berry, D., et al. 2016, in *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, Vol. 9914, Proceedings of the SPIE, 991403, doi: [10.1117/12.2231943](https://doi.org/10.1117/12.2231943)
- Gandilo, N. N., Ade, P. A. R., Angilè, F. E., et al. 2016, *ApJ*, 824, 84, doi: [10.3847/0004-637X/824/2/84](https://doi.org/10.3847/0004-637X/824/2/84)
- Goldreich, P., & Kylafis, N. D. 1981, *ApJL*, 243, L75, doi: [10.1086/183446](https://doi.org/10.1086/183446)
- . 1982, *ApJ*, 253, 606, doi: [10.1086/159663](https://doi.org/10.1086/159663)
- Goodman, A. A., Jones, T. J., Lada, E. A., & Myers, P. C. 1995, *ApJ*, 448, 748, doi: [10.1086/176003](https://doi.org/10.1086/176003)
- Guillet, V., Fanciullo, L., Verstraete, L., et al. 2018, *A&A*, 610, A16, doi: [10.1051/0004-6361/201630271](https://doi.org/10.1051/0004-6361/201630271)
- Hennebelle, P., & Inutsuka, S.-i. 2019, *Frontiers in Astronomy and Space Sciences*, 6, 5, doi: [10.3389/fspas.2019.00005](https://doi.org/10.3389/fspas.2019.00005)
- Hildebrand, R. H. 1983, *QJRAS*, 24, 267
- Hildebrand, R. H., Dotson, J. L., Dowell, C. D., Schleuning, D. A., & Vaillancourt, J. E. 1999, *ApJ*, 516, 834, doi: [10.1086/307142](https://doi.org/10.1086/307142)
- Hildebrand, R. H., Kirby, L., Dotson, J. L., Houde, M., & Vaillancourt, J. E. 2009, *ApJ*, 696, 567, doi: [10.1088/0004-637X/696/1/567](https://doi.org/10.1088/0004-637X/696/1/567)
- Hoang, T., Cho, J., & Lazarian, A. 2018, *ApJ*, 852, 129, doi: [10.3847/1538-4357/aa9edc](https://doi.org/10.3847/1538-4357/aa9edc)
- Hoang, T., & Lazarian, A. 2016, *ApJ*, 831, 159, doi: [10.3847/0004-637X/831/2/159](https://doi.org/10.3847/0004-637X/831/2/159)
- Holland, W. S., Bintley, D., Chapin, E. L., et al. 2013, *Monthly Notices of the Royal Astronomical Society*, 430, 2513, doi: [10.1093/mnras/sts612](https://doi.org/10.1093/mnras/sts612)
- Hull, C. L. H., Plambeck, R. L., Kwon, W., et al. 2014, *ApJS*, 213, 13, doi: [10.1088/0067-0049/213/1/13](https://doi.org/10.1088/0067-0049/213/1/13)
- Köhler, M., Ysard, N., & Jones, A. P. 2015, *A&A*, 579, A15, doi: [10.1051/0004-6361/201525646](https://doi.org/10.1051/0004-6361/201525646)
- Kwon, J., Doi, Y., Tamura, M., et al. 2018, *The Astrophysical Journal*, 859, 4, doi: [10.3847/1538-4357/aabd82](https://doi.org/10.3847/1538-4357/aabd82)
- Lazarian, A. 2007, *JQSRT*, 106, 225, doi: [10.1016/j.jqsrt.2007.01.038](https://doi.org/10.1016/j.jqsrt.2007.01.038)
- Lazarian, A., & Hoang, T. 2007a, *MNRAS*, 378, 910, doi: [10.1111/j.1365-2966.2007.11817.x](https://doi.org/10.1111/j.1365-2966.2007.11817.x)
- . 2007b, *ApJL*, 669, L77, doi: [10.1086/523849](https://doi.org/10.1086/523849)

- , 2008, *ApJL*, 676, L25, doi: [10.1086/586706](https://doi.org/10.1086/586706)
- Mairs, S., Dempsey, J. T., Bell, G. S., et al. 2021, *AJ*, 162, 191, doi: [10.3847/1538-3881/ac18bf](https://doi.org/10.3847/1538-3881/ac18bf)
- Michail, J. M., Ashton, P. C., Berthoud, M. G., et al. 2021, *ApJ*, 907, 46, doi: [10.3847/1538-4357/abd090](https://doi.org/10.3847/1538-4357/abd090)
- Montier, L., Plaszczyński, S., Levrier, F., et al. 2015, *A&A*, 574, A136, doi: [10.1051/0004-6361/201424451](https://doi.org/10.1051/0004-6361/201424451)
- Pattle, K., Fissel, L., Tahani, M., Liu, T., & Ntormousi, E. 2023, in *Astronomical Society of the Pacific Conference Series*, Vol. 534, *Protostars and Planets VII*, ed. S. Inutsuka, Y. Aikawa, T. Muto, K. Tomida, & M. Tamura, 193, doi: [10.48550/arXiv.2203.11179](https://doi.org/10.48550/arXiv.2203.11179)
- Pattle, K., Fissel, L. M., & Pineda, J. E. 2024, *Annual Review of Astronomy and Astrophysics*, 62, in press, doi: [10.1146/annurev-astro-022823-040828](https://doi.org/10.1146/annurev-astro-022823-040828)
- Pattle, K., Ward-Thompson, D., & Fissel, L. M. 2022, *arXiv e-prints*, arXiv:2203.11179, <https://arxiv.org/abs/2203.11179>
- Pattle, K., Ward-Thompson, D., Berry, D., et al. 2017, *The Astrophysical Journal*, 846, 122, doi: [10.3847/1538-4357/aa80e5](https://doi.org/10.3847/1538-4357/aa80e5)
- Pattle, K., Lai, S.-P., Di Francesco, J., et al. 2021, *ApJ*, 907, 88, doi: [10.3847/1538-4357/abcc6c](https://doi.org/10.3847/1538-4357/abcc6c)
- Planck Collaboration. 2020, *A&A*, 641, A12, doi: [10.1051/0004-6361/201833885](https://doi.org/10.1051/0004-6361/201833885)
- Planck Collaboration, Ade, P. A. R., Aghanim, N., et al. 2015, *Astronomy & Astrophysics*, 576, A104, doi: [10.1051/0004-6361/201424082](https://doi.org/10.1051/0004-6361/201424082)
- Plaszczyński, S., Montier, L., Levrier, F., & Tristram, M. 2014, *MNRAS*, 439, 4048, doi: [10.1093/mnras/stu270](https://doi.org/10.1093/mnras/stu270)
- Robitaille, T. P., Whitney, B. A., Indebetouw, R., Wood, K., & Denzmore, P. 2006, *ApJS*, 167, 256, doi: [10.1086/508424](https://doi.org/10.1086/508424)
- Serkowski, K., Mathewson, D. S., & Ford, V. L. 1975, *ApJ*, 196, 261, doi: [10.1086/153410](https://doi.org/10.1086/153410)
- Shariff, J. A., Ade, P. A. R., Angilè, F. E., et al. 2019, *The Astrophysical Journal*, 872, 197, doi: [10.3847/1538-4357/aaff5f](https://doi.org/10.3847/1538-4357/aaff5f)
- Soam, A., Pattle, K., Ward-Thompson, D., et al. 2018, *The Astrophysical Journal*, 861, 65, doi: [10.3847/1538-4357/aac4a6](https://doi.org/10.3847/1538-4357/aac4a6)
- Vaillancourt, J. E., & Matthews, B. C. 2012, *ApJS*, 201, 13, doi: [10.1088/0067-0049/201/2/13](https://doi.org/10.1088/0067-0049/201/2/13)
- Vaillancourt, J. E., Dowell, C. D., Hildebrand, R. H., et al. 2008, *ApJL*, 679, L25, doi: [10.1086/589152](https://doi.org/10.1086/589152)
- Ward-Thompson, D., Pattle, K., Bastien, P., et al. 2017, *The Astrophysical Journal*, 842, 66, doi: [10.3847/1538-4357/aa70a0](https://doi.org/10.3847/1538-4357/aa70a0)
- Whitney, B. A., Wood, K., Bjorkman, J. E., & Cohen, M. 2003, *ApJ*, 598, 1079, doi: [10.1086/379068](https://doi.org/10.1086/379068)
- Whittet, D. C. B., Hough, J. H., Lazarian, A., & Hoang, T. 2008, *ApJ*, 674, 304, doi: [10.1086/525040](https://doi.org/10.1086/525040)
- Ysard, N., Koehler, M., Jimenez-Serra, I., Jones, A. P., & Verstraete, L. 2019, *A&A*, 631, A88, doi: [10.1051/0004-6361/201936089](https://doi.org/10.1051/0004-6361/201936089)
- Ysard, N., Jones, A. P., Guillet, V., et al. 2024, *A&A*, 684, A34, doi: [10.1051/0004-6361/202348391](https://doi.org/10.1051/0004-6361/202348391)